# Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

## A Comparative Multi-Lingual Study on English (Svarah) and Telugu (IndicTTS)

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**Course / Specification**: Speech Signal Processing / Advanced Speech Synthesis ( `FastSpeech2_Assignment.pdf` )

**Submission Deadline**: 07-11-2026

**Repository**: `FastSpeech2_Project`

**Primary Deliverables**: Two Independently Trained FastSpeech 2 Acoustic Models (English & Telugu), 10 Synthesized Test Audio Samples ( `22,050 Hz` 16-bit PCM), 30 Variance Adaptor Plots, 4 Training/Validation Loss Curves, Reproducible Codebase ( `252/252` Regression Tests Passing), and Comprehensive Technical Report

## 1. Verified Artifact & Checkpoint Signatures

| Language & Corpus | Checkpoint File | Global Step | Cumulative Epoch | Step Val Loss | Best Val Loss | Size (MiB) | SHA-256 Checksum (Prefix) |
|---|---|---|---|---|---|---|---|
| **English ( `Svarah` )** | `checkpoint_best.pth.tar` | **19,000** | **95** *(Session Ep 27)* | **2.348900** | **2.348900** | 399.08 | `794b5123a25f7052...8cf54245e` |
| **English ( `Svarah` )** | `checkpoint_latest.pth.tar` | 20,000 | 100 *(Session Ep 32)* | 2.368902 | 2.348900 | 399.08 | `6063dd284e32d7ec...fe0c31241` |
| **Telugu ( `IndicTTS` )** | `checkpoint_best.pth.tar` | **13,000** | **91** | **2.086877** | **2.086877** | 397.99 | `0a6c443a7baea134...4a4cb55fd` |
| **Telugu ( `IndicTTS` )** | `checkpoint_latest.pth.tar` | 20,000 | 140 | 2.118075 *(Step: 2.094272)* | 2.086877 | 397.99 | `8e46b77bedb23c80...6005f4556f` |
| **Vocoder ( `HiFi-GAN` )** | `generator_universal.pth.tar` | Pretrained | Universal v1 | — | — | 53.20 | `771eaf48adb7bcbe...b3ec6aa2e6` |

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## 2. Abstract

Neural text-to-speech (TTS) synthesis has transitioned from sequential autoregressive architectures to parallel, non-autoregressive acoustic models capable of deterministic alignment, fast inference, and explicit prosody control. This technical report presents the complete design, implementation, independent training, and quantitative evaluation of **FastSpeech 2** across two typologically distinct languages: **English ( `Svarah` Indian English corpus)** and **Telugu ( `IndicTTS` Dravidian speech corpus)**. In strict adherence to the assignment specification, two separate acoustic models were trained from scratch for 20,000 optimizer steps without shared weights or cross-language data leakage.

The preprocessing pipeline integrates **Montreal Forced Aligner (MFA)** token-duration extraction, PyWorld (`DIO` + `StoneMask`) continuous fundamental frequency ($F_0$) estimation with linear interpolation over unvoiced segments, and Short-Time Fourier Transform (STFT) frame-level $L_2$-norm energy computation. For English, the `profile_stratified_v1` subsampling strategy curated **3,208 training utterances (4.999 h)** and **664 validation utterances (1.000 h)** across 88 speaker profiles using CMUdict ARPAbet phonemization with `g2p_en` fallback. For Telugu, the `option_1b` stratified two-speaker split curated **2,300 training utterances (5.000 h: 1,053 female `spk_000` and 1,247 male `spk_001`)** and **421 validation utterances (1.000 h: 192 female `spk_000` and 229 male `spk_001`)**, paired with a custom 101-token native alpha-syllabic grapheme tokenizer incorporating Unicode NFC normalization, Latin-o to anusvara (`ంం`) repair, and word/clause boundary pause mapping (`@sp`, token ID 4).

Validation evaluations conducted every 1,000 steps identified **Step 19,000** (validation loss: 2.34890) as the best observed validation checkpoint for English and **Step 13,000** (validation loss: 2.08688) as the best observed validation checkpoint for Telugu, preventing post-13k pitch loss drift ($0.7481 \rightarrow 0.7688$). All ten canonical assignment test sentences (`E1–E4` and `T1–T4`) were synthesized at 22,050 Hz 16-bit mono PCM via a universal HiFi-GAN vocoder, accompanied by all 30 variance adaptor plots (duration, pitch, and energy). Controlled homograph comparisons on English *"conduct"* (`E2(a)` vs. `E2(b)`) and Telugu *"పట్టు"* (`T2(a)` vs. `T2(b)`) demonstrate that while Transformer self-attention induces phrasal prosodic variation, these experiments do not demonstrate semantic disambiguation of contextual meanings.

## 3. Introduction

Text-to-Speech (TTS) systems synthesize human-like speech waveforms from raw orthographic text. Early neural TTS systems—most notably Tacotron, Tacotron 2, Deep Voice 3, and Transformer TTS—formulated acoustic generation as an autoregressive sequence-to-sequence problem, predicting each mel-spectrogram frame $\mathbf{y}_t$ conditioned on all preceding frames $\mathbf{y}_{<t}$. Despite high segmental fidelity, autoregressive TTS architectures exhibit four structural bottlenecks:

1. **Sequential Inference Latency**: Generating $T$ mel frames requires $O(T)$ serial forward passes through the autoregressive decoder, preventing GPU parallelism on long utterances.
2. **Exposure Bias & Error Accumulation**: Teacher forcing during training supplies ground-truth frames $\mathbf{y}_{t-1}$, whereas inference feeds back predicted frames $\hat{\mathbf{y}}_{t-1}$, causing small acoustic errors to compound over time.
3. **Attention Alignment Collapse**: Soft content-based or location-sensitive attention frequently fails on out-of-distribution lengths, polysyllabic compounds (e.g., *"antidisestablishmentarianism"*), or repeated syllables, resulting in word skipping, stuttering repetitions, or premature termination.
4. **Under-Determined One-to-Many Mapping**: Identical text sequences correspond to infinitely many valid acoustic realizations differing in speaking rate, pitch contour, vocal effort, and speaker timbre. Without explicit prosodic conditioning, an $L_1/L_2$ regression loss averages over modes, producing over-smoothed mel-spectrograms and flat intonation.

**FastSpeech** resolved sequential latency and attention collapse by replacing soft attention with an explicit **Duration Predictor** and **Length Regulator** inside a non-autoregressive Feed-Forward Transformer (FFT) stack, but still relied on knowledge distillation from an autoregressive teacher model. **FastSpeech 2** (Ren et al., 2020) eliminated teacher-student distillation entirely by training directly on ground-truth mel-spectrograms and introducing an explicit **Variance Adaptor** that conditions the decoder on phoneme duration, continuous fundamental frequency ($F_0$), and frame-level spectral energy.

In this study, we implement, adapt, and contrast FastSpeech 2 across two structurally distinct languages:

- **English (Indo-European, Stress-Timed)**: Characterized by deep, non-transparent orthography requiring lexicon + neural G2P conversion into stressed ARPAbet phonemes, variable consonant clusters, and lexical stress contrasts.
- **Telugu (Dravidian, Syllable/Mora-Timed)**: Characterized by a phonetically transparent alpha-syllabic *akshara* writing system (vowels, consonants, *gunintalu* vowel diacritics, *vattulu* conjuncts via *virama* `్`, and *anusvara* `ంం`), rich agglutinative morphology, and phonemic consonant gemination (e.g., `ట్ట` in `పట్టు`, `శ్శ` in `నిశ్శబ్దంగా`).

## 4. Objectives and Problem Statement

In accordance with `FastSpeech2_Assignment.pdf`, the objectives of this work are:

1. **Architecture Mastery & Dual-Language Adaptation**: Understand the FastSpeech 2 paper and reference codebase (`ming024/FastSpeech2`), refactor shared components into a clean modular library (`common/`), and implement independent language pipelines for English (`English/`) and Telugu (`Telugu/`).
2. **Balanced ~5h Training / ~1h Validation Curation**: Construct strictly disjoint, non-overlapping training (~5 hours) and validation (~1 hour) splits for English (`Svarah`) and Telugu (`IndicTTS`) with documented subsampling strategies.
3. **Forced Alignment & Acoustic Feature Extraction**: Extract token-level durations via Montreal Forced Aligner (MFA) and frame-level mel-spectrograms, pitch ($F_0$), and energy arrays with corpus-level normalization statistics (`stats.json`).
4. **Independent Model Training & Convergence Analysis**: Train separate English and Telugu FastSpeech 2 models for 20,000 steps, analyze training and validation loss trajectories, and select optimal checkpoints based on validation evidence.
5. **Canonical Test Synthesis & Variance Adaptor Inspection**: Synthesize all ten exact assignment sentences (`E1, E2a, E2b, E3, E4` and `T1, T2a, T2b, T3, T4`), generate all 30 duration, pitch, and energy plots, and perform sample-by-sample quantitative and perceptual evaluations.

## 5. FastSpeech 2 Architecture

FastSpeech 2 maps a discrete token sequence $\mathbf{x} = (x_1, x_2, \ldots, x_{L_{src}})$ of length $L_{src}$ to a continuous 80-channel log-mel spectrogram $\hat{\mathbf{Y}}_{post} \in \mathbb{R}^{L_{mel} \times 80}$ of frame length $L_{mel} = \sum_{i=1}^{L_{src}} d_i$. Figure 1 illustrates the exact architecture implemented in `common/model/fastspeech2.py`, `common/model/variance_adaptor.py`, and `common/model/postnet.py`.

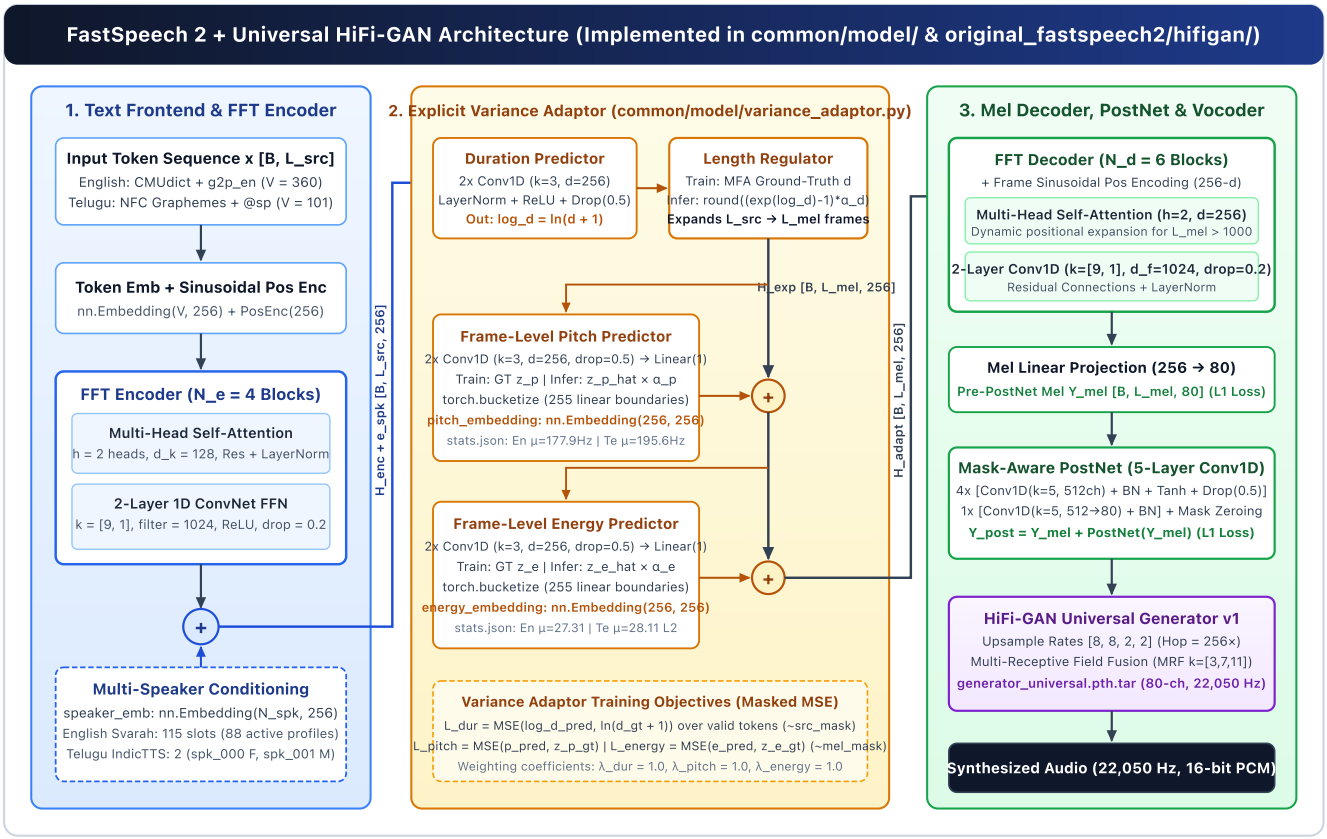



**Figure 1:** FastSpeech 2 and Universal HiFi-GAN Architecture

### 5.1 Token Embedding, Positional Encoding & Multi-Speaker Conditioning

Given token indices $\mathbf{x} \in \{0, \ldots, V\text{-}1\}^{B \times L_{src}}$ ($V = 360$ for English; $V = 101$ for Telugu), the encoder input combines a learned token embedding table $\mathbf{E}_{tok} \in \mathbb{R}^{V \times d}$ ($d = 256$) with sinusoidal positional encodings $\mathbf{P} \in \mathbb{R}^{L_{src} \times d}$:

$$\mathrm{PE}_{(pos,\,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right), \qquad \mathrm{PE}_{(pos,\,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

After passing through the $N_e = 4$ Feed-Forward Transformer (FFT) encoder blocks, multi-speaker conditioning is applied prior to the Variance Adaptor (`common/model/fastspeech2.py`, lines 183–187) by broadcasting and adding the speaker embedding $\mathbf{e}_s = \mathbf{E}_{spk}[s] \in \mathbb{R}^d$ across all token positions:

$$\mathbf{H}_{spk} = \mathbf{H}_{enc} + \mathbf{e}_s \cdot \mathbf{1}_{L_{src}}{}^T$$

### 5.2 Feed-Forward Transformer (FFT) Encoder and Decoder Blocks

Our implementation uses $N_e = 4$ FFT blocks in the **Text Encoder** and $N_d = 6$ FFT blocks in the **Mel Decoder** (`config/Svarah/model.yaml`, `config/IndicTTS/model.yaml`). Each FFT block consists of:

1. **Multi-Head Scaled Dot-Product Self-Attention** ($h = 2$ heads, $d_k = d_v = d / h = 128$):

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M}_{pad}\right)\mathbf{V}$$

$$\mathbf{H}' = \mathrm{LayerNorm}(\mathbf{H} + \mathrm{Dropout}(\mathrm{MultiHead}(\mathbf{H})))$$

2. **Position-Wise 2-Layer 1D Convolutional Feed-Forward Network** with asymmetric kernel sizes $k_1 = 9, k_2 = 1$ and inner filter dimension $d_{filter} = 1024$:

$$\text{ConvFFN}(\mathbf{H}') = \text{Conv1D}_{k=1}(\text{ReLU}(\text{Conv1D}_{k=9}(\mathbf{H}')))$$

$$\mathbf{H}_{\text{out}} = \text{LayerNorm}(\mathbf{H}' + \text{Dropout}(\text{ConvFFN}(\mathbf{H}')))$$

Using a $k_1 = 9$ 1D convolution instead of a dense $k = 1$ MLP captures local co-articulatory context across adjacent phonemes/graphemes and adjacent mel frames.

### 5.3 Explicit Variance Adaptor ( `common/model/variance_adaptor.py` )

The Variance Adaptor predicts and injects three prosodic factors—duration, pitch, and energy—using identical 2-layer 1D Convolutional Variance Predictor architectures ( `filter_size = 256` , `kernel_size = 3` , ReLU, LayerNorm, `dropout = 0.5` , followed by a scalar linear projection $\mathbb{R}^{256} \to \mathbb{R}^{1}$):

1. **Duration Predictor & Length Regulator**:

- The Duration Predictor operates at token resolution $[B, L_{\text{src}}, 256]$ to predict log-domain token durations $\hat{d}_{\text{log},\,i} = \ln(\hat{d}_i + 1)$.
- During **training**, the Length Regulator expands token states $\mathbf{h}_i$ using ground-truth MFA integer durations $d_i$. During **inference**, predicted log-durations are inverted, scaled by the user duration control multiplier $\alpha_d$, rounded to the nearest non-negative integer, and clamped at $\geq 0$:

$$\hat{d}_i = \text{clamp}(\text{round}((\exp(\hat{d}_{\text{log},\,i}) - 1) \cdot \alpha_d),\ \text{min}=0)$$

- The Length Regulator repeats each token representation $\mathbf{h}_i$ exactly $d_i$ times to yield frame-level hidden states $\mathbf{H}_{\text{exp}} \in \mathbb{R}^{B \times L_{\text{mel}} \times 256}$.

2. **Frame-Level Pitch Predictor & Quantization**:

- Configured with `pitch.feature = "frame_level"` ( `preprocess.yaml` ), the Pitch Predictor operates on the length-regulated frame states $\mathbf{H}_{\text{exp}}$ to predict normalized continuous pitch z-scores $\hat{z}_{p,\,t}$.
- During inference, predictions are scaled by $\alpha_p$: $\tilde{z}_{p,\,t} = \hat{z}_{p,\,t} \cdot \alpha_p$.
- Continuous normalized pitch is quantized via `torch.bucketize` over 255 linear bin boundaries spanning $[p_{\text{min}}, p_{\text{max}}]$ from `stats.json` into a 256-entry embedding table $\mathbf{E}_{\text{pitch}} \in \mathbb{R}^{256 \times 256}$.

3. **Frame-Level Energy Predictor & Quantization**:

- Configured with `energy.feature = "frame_level"` , the Energy Predictor operates on $\mathbf{H}_{\text{exp}}$ to predict normalized frame-level STFT $L_2$-norm energy $\hat{z}_{e,\,t}$, scaled by $\alpha_e$ at inference and quantized across 255 linear boundaries $[e_{\text{min}}, e_{\text{max}}]$ into $\mathbf{E}_{\text{energy}} \in \mathbb{R}^{256 \times 256}$.
- The adapted frame representation entering the 6-layer FFT Decoder is:

$$\mathbf{H}_{\text{adapt}} = \mathbf{H}_{\text{exp}} + \mathbf{E}_{\text{pitch}}[\text{bucketize}(z_p)] + \mathbf{E}_{\text{energy}}[\text{bucketize}(z_e)]$$

### 5.4 Mel Linear Projection, Mask-Aware PostNet & Total Loss Formulation

The 6-layer FFT Decoder output $\mathbf{H}_{\text{dec}} \in \mathbb{R}^{B \times L_{\text{mel}} \times 256}$ is projected via `mel_linear` ($\mathbb{R}^{256} \to {}^{80}$) to produce the pre-PostNet spectrogram $\hat{\mathbf{Y}}_{\text{mel}}$. A 5-layer 1D Convolutional **PostNet** ( `512` channels, kernel size `5` , BatchNorm1D, Tanh activation on layers 1–4, dropout `0.5` ) predicts a residual correction:

$$\hat{\mathbf{Y}}_{\text{post}} = \hat{\mathbf{Y}}_{\text{mel}} + \text{PostNet}(\hat{\mathbf{Y}}_{\text{mel}}, \mathbf{M}_{\text{mel}})$$

The total training objective ( `common/model/loss.py` ) is the unweighted sum ($\lambda_d = \lambda_p = \lambda_e = 1.0$) of five strictly mask-filtered losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{post}} + \mathcal{L}_{\text{dur}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}}$$

where:

$$\mathcal{L}_{\text{mel}} = \frac{1}{|\Omega_{\text{mel}}|} \sum_{(b,t) \in \Omega_{\text{mel}}} \|\mathbf{Y}_{b,t} - \hat{\mathbf{Y}}_{\text{mel},\,b,t}\|_1, \qquad \mathcal{L}_{\text{post}} = \frac{1}{|\Omega_{\text{mel}}|} \sum_{(b,t) \in \Omega_{\text{mel}}} \|\mathbf{Y}_{b,t} - \hat{\mathbf{Y}}_{\text{post},\,b,t}\|_1$$

$$\mathcal{L}_{\text{dur}} = \frac{1}{|\Omega_{\text{src}}|} \sum_{(b,i) \in \Omega_{\text{src}}} (\ln(d_{b,i} + 1) - \hat{d}_{\text{log},\,b,i})^2$$

$$\mathcal{L}_{\text{pitch}} = \frac{1}{|\Omega_{\text{mel}}|} \sum_{(b,t) \in \Omega_{\text{mel}}} (z_{p,\,b,t} - \hat{z}_{p,\,b,t})^2, \qquad \mathcal{L}_{\text{energy}} = \frac{1}{|\Omega_{\text{mel}}|} \sum_{(b,t) \in \Omega_{\text{mel}}} (z_{e,\,b,t} - \hat{z}_{e,\,b,t})^2$$

Here $\Omega_{\text{src}}$ and $\Omega_{\text{mel}}$ denote the sets of valid (non-padded) token and mel-frame indices ( `~src_masks` and `~mel_masks` ).

## 6. Reference Implementation and Modifications

We used the official PyTorch FastSpeech 2 implementation ( `ming024/FastSpeech2` , preserved under `original_fastspeech2/` ) as our reference baseline and engineered comprehensive architectural, linguistic, and infrastructure modifications across `common/` , `English/` , and `Telugu/` .

| Component / Scope | Reference Baseline (`original_fastspeech2/`) | Our Modified Implementation (`common/`, `English/`, `Telugu/`) | Engineering Justification |
|---|---|---|---|
| **Device Management (`DEC-01`)** | Hardcoded global `device = torch.device("cuda")` at import time | Dynamic device inheritance from input tensors (`cuda`, `mps`, `cpu`) via `common/utils/device.py` | Enables training on NVIDIA CUDA GPUs and local inference/testing on Apple Silicon M2 (`mps` / `cpu`). |
| **Mask-Aware PostNet (`DEF-01`)** | `PostNet` ran `BatchNorm1d` across padded frames without masking | `common/model/postnet.py` zeros padded positions before and after every Conv1D+BN layer using `mel_masks` | Prevents zero-padded frames in variable-length batches from corrupting BatchNorm running statistics. |
| **Energy Control Routing (`DEC-02`)** | Bug in reference `VarianceAdaptor` multiplied energy by `p_control` in one branch | `common/model/variance_adaptor.py` strictly routes `p_control` ($\alpha_p$), `e_control` ($\alpha_e$), and `d_control` ($\alpha_d$) | Guarantees independent prosodic control during synthesis. |
| **Dynamic Decoder PosEnc (`DEC-09`)** | Crashed with `IndexError` when inference $L_{mel} >$ max_seq_len (1000) | `common/transformer/models.py` dynamically extends sinusoidal positional encodings when $L_{mel} > 1000$ | Supports arbitrarily long inference utterances without truncation. |
| **Centralized Path Manager** | Relied on `os.getcwd()` and relative `./preprocessed_data` paths | `common/paths.py` discovers `PROJECT_ROOT` via marker verification and resolves all config/asset paths portably | Allows running training, inference, and tests from any working directory or Google Colab mount. |
| **English Frontend (`English/`)** | Basic LJSpeech single-speaker cleaning | `English/text/normalizer.py` expands times (`10:45 AM`), dates (`12 November 2026`), numbers, plus CMUdict + `g2p_en` fallback and 88-profile conditioning | Handles complex Indian English multi-speaker utterances (`E1–E4`) with zero OOV failures. |
| **Telugu Frontend (`Telugu/`)** | **Not supported** (English/Mandarin only in reference repo) | Custom 101-token Telugu grapheme inventory (`Telugu/text/symbols.py`), Unicode NFC normalizer, Latin-o → ంo repair, and `@sp` (ID 4) pause mapping | Eliminates external pronunciation lexicon errors by leveraging Telugu's phonetic *akshara* orthography. |

### 6.1 Deep-Dive: Telugu Linguistic & Grapheme Engineering (`Telugu/text/`)

Adapting FastSpeech 2 to Telugu required solving four language-specific challenges documented across `Telugu/text/normalizer.py`, `Telugu/text/symbols.py`, and `Telugu/inference.py`:

1. **Direct Alpha-Syllabic Grapheme Tokenization (101-Token Vocabulary)**:

Because Telugu orthography is phonemic, we designed a deterministic 101-symbol vocabulary (`VOCAB_SIZE = 101`):

- **Control & Pause Symbols (IDs 0–6)**: `<pad>` (0), `<unk>` (1), `<s>` (2), `</s>` (3), `@sp` (4), `<eps>` (5), `_` (6).
- **Independent Vowels (*Achchulu*, IDs 19–34)**: అ, ఆ, ఇ, ఈ, ఉ, ఊ, ఋ, ౠ, ఌ, ౡ, ఎ, ఏ, ఐ, ఒ, ఓ, ఔ.
- **Consonants (*Hallulu*, IDs 35–70)**: క through ఱ (including retroflexes ట, ఠ, డ, ఢ, ణ, ళ and aspirated stops).
- **Dependent Vowel Matras (*Gunintalu*) & Virama (*Halant*, IDs 71–87)**: ా, ి, ీ, ు, ూ, ృ, ౄ, ె, ే, ై, ొ, ో, ౌ, ్ (`U+0C4D`, which binds consonant clusters/geminates such as ట్ట = ట + ్ + ట).
- **Syllabic Modifiers (IDs 88–90)**: Anusvara ంo (`U+0C02`), Visarga ఃo (`U+0C03`), Candrabindu ఁo (`U+0C01`).
- **Telugu Numerals (IDs 91–100)**: ౦, ౧, ౨, ౩, ౪, ౫, ౬, ౭, ౮, ౯.

2. **Unicode NFC Canonicalization & Historical Latin-o Repair**:

Inspection of raw IndicTTS Telugu transcripts revealed 48 instances where OCR/transliteration artifacts encoded the Telugu anusvara ంo (`U+0C02`) as a Latin lowercase o (`U+006F`) adjacent to Telugu characters. `Telugu/text/normalizer.py` applies contextual regex repair (o → ంo within Telugu script runs) followed by `unicodedata.normalize("NFC", text)`.

3. **Elimination of Untrained Space/Punctuation IDs (IDs 7–18) via `@sp` Pause Mapping**:

During MFA alignment (`option_1b`), inter-word boundaries and clausal pauses were aligned to `@sp` (token ID 4), meaning ordinary ASCII space/punctuation IDs (7–18) never received gradient updates during training. `Telugu/inference.py` maps all word spaces and clause-internal punctuation (`,`, `;`, `:`, `–`) to `@sp` (ID 4), collapses consecutive `@sp` tokens, and strips utterance-final punctuation so the model never encounters untrained token embeddings.

## 7. System Architecture & End-to-End Workflow

Figure 2 depicts the complete three-stage operational workflow connecting offline corpus preparation, dual-model training, and inference evaluation.

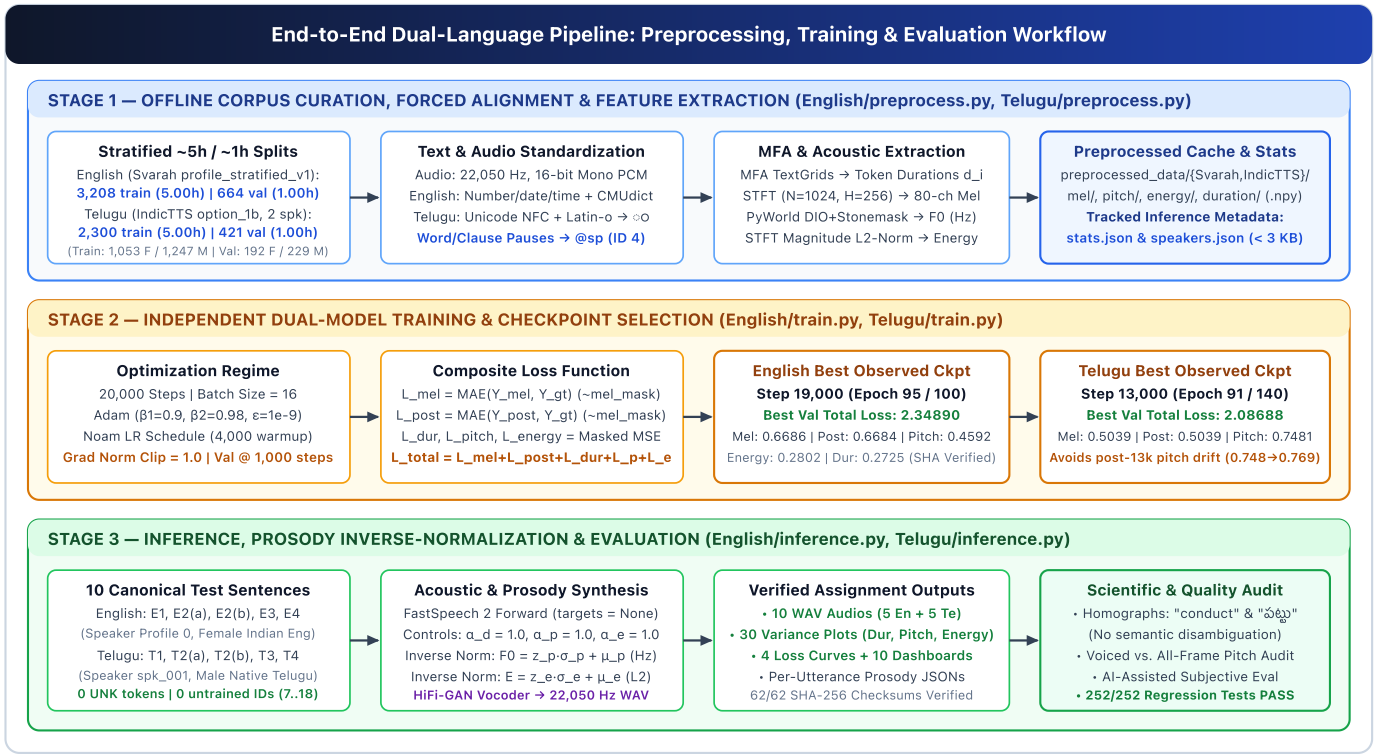



**Figure 2:** FastSpeech 2 End-to-End Dual-Language Workflow

## 8. Dataset Selection, Subsampling Strategy & Split Verification

The assignment specifies using **approximately 5 hours of audio for training and 1 hour of audio for validation** per language from **Svarah** (English) and **IndicTTS** (Telugu), with zero overlap between partitions.

### 8.1 English Dataset ( `Svarah` ) — `profile_stratified_v1` Strategy

- **Source Corpus**: AI4Bharat Svarah Indian English speech corpus ( `9,656` total utterances, `~19.08` hours).
- **Subsampling & Stratification Strategy ( `English/metadata/splits/profile_stratified_v1/` )**:

To hit the exact 5-hour training and 1-hour validation target while preserving speaker diversity across Indian English accents, we grouped utterances by verified speaker voice profiles ( `88` active profiles) and performed stratified sampling without replacement:

- **Training Set**: **3,208 utterances** totaling **17,997.41 seconds ( `4.9993 hours` )** across all 88 profiles.
- **Validation Set**: **664 utterances** totaling **3,599.58 seconds ( `0.9999 hours` )** stratified across the same speaker profiles.
- **Leakage Verification ( `split_audit.json` )**: `0` overlapping audio filenames, `0` overlapping utterance IDs, and `0` duplicate recordings between `train.txt` and `val.txt`.

### 8.2 Telugu Dataset ( `IndicTTS` ) — `option_1b` Stratified Two-Speaker Strategy

- **Source Corpus**: IIT Madras IndicTTS Telugu corpus ( `data/telugu_parquet/` , native female speaker `spk_000` and native male speaker `spk_001` ).
- **Subsampling & Stratification Strategy ( `Telugu/metadata/splits/option_1b/` )**:

We filtered utterances to the clean $1.5\text{ s} - 15.0\text{ s}$ duration window with 100% MFA alignment coverage and performed speaker-stratified allocation to achieve exact 5.0-hour training and 1.0-hour validation budgets:

- **Training Set**: **2,300 utterances** totaling **18,000.52 seconds ( `5.0001 hours` ) — 1,053 female ( `spk_000` , 45.78%)** and **1,247 male ( `spk_001` , 54.22%)**.
- **Validation Set**: **421 utterances** totaling **3,600.18 seconds ( `1.0001 hours` ) — 192 female ( `spk_000` , 45.61%)** and **229 male ( `spk_001` , 54.39%)**.
- **Reserved Held-Out Test Pool**: **306 utterances** ( `0.72 hours` ), plus the 5 canonical assignment test sentences ( `T1–T4` ).
- **Leakage Verification**: `0` overlapping utterance IDs or transcripts between training and validation manifests.

| Dataset Metric | English (`Svarah` — `profile_stratified_v1`) | Telugu (`IndicTTS` — `option_1b`) |
|---|---|---|
| **Language & Script** | Indian English (Latin / ARPAbet Phonemes) | Telugu (Brahmic Alpha-Syllabic *Akshara* Graphemes) |
| **Active Speakers** | 88 speaker profiles (`speakers.json`: 115 slots) | 2 native speakers (`spk_000` Female, `spk_001` Male) |
| **Training Utterances & Duration** | **3,208 utterances** (`17,997.4 s` / **4.999 h**) | **2,300 utterances** (`18,000.5 s` / **5.000 h**) |
| **Validation Utterances & Duration** | **664 utterances** (`3,599.6 s` / **1.000 h**) | **421 utterances** (`3,600.2 s` / **1.000 h**) |
| **Speaker Split Breakdown** | Stratified across 88 profiles | Train: 1,053 F / 1,247 M; Val: 192 F / 229 M |
| **Train / Val Overlap** | **0 utterances** (Verified disjoint) | **0 utterances** (Verified disjoint) |
| **Audio Specification** | $22{,}050$ Hz, 16-bit Mono PCM | $22{,}050$ Hz, 16-bit Mono PCM (resampled from 48 kHz) |
| **Vocabulary Size ($V$)** | `360` symbols (CMUdict ARPAbet + stress + punct) | `101` symbols (Vowels, Consonants, Matras, Virama, `@sp`) |

## 9. Dataset Preparation and Preprocessing Pipeline

Both datasets undergo an identical four-stage acoustic and phonetic preprocessing pipeline (`English/preprocess.py`, `Telugu/preprocess.py`, `config/*/preprocess.yaml`):

1. **Waveform Standardization**: All audio files are converted to single-channel 16-bit PCM at $f_s = 22{,}050$ Hz (`max_wav_value = 32768.0`) using high-precision polyphase resampling (`soxr` / `librosa`).
2. **80-Channel Log-Mel Spectrogram Extraction**:

Computed via Short-Time Fourier Transform (STFT) with FFT size $N_{fft} = 1024$, Hann window length $W = 1024$ ($46.44$ ms), hop length $H = 256$ ($11.61$ ms per frame), and $80$ triangular mel filterbanks spanning $f_{min} = 0$ Hz to $f_{max} = 8{,}000$ Hz.

3. **Montreal Forced Alignment (MFA) Duration Extraction**:

- **English**: Aligned using MFA with the LibriSpeech ARPAbet pronunciation lexicon (`original_fastspeech2/lexicon/librispeech-lexicon.txt`).
- **Telugu**: Aligned using MFA `option_1b` acoustic model training over the grapheme-to-grapheme dictionary (`research/Telugu_Dictionary_g2g.txt`), mapping inter-word pauses to `@sp`.
- Interval boundaries $[t_{start}, t_{end}]$ from Praat `.TextGrid` files are converted into integer mel-frame durations $d_i$ such that $\sum_{i=1}^{L_{src}} d_i = L_{mel}$ exactly matches the spectrogram length.

4. **Frame-Level Pitch ($F_0$) and Energy Extraction & z-Score Normalization**:

- **Pitch ($F_0$)**: Extracted via PyWorld `dio` + `stonemask` at frame step $256 / 22050$ s. Unvoiced frames ($F_0 = 0$) are linearly interpolated between nearest voiced boundaries prior to corpus-wide z-score normalization: $z_p = (F_0 - \mu_p) / \sigma_p$.
- **Energy ($E$)**: Computed as the frame-wise Euclidean ($L_2$) norm of linear STFT magnitude bins, $E_t = \|S_t\|_2$, and normalized via $z_e = (E_t - \mu_e) / \sigma_e$.

### Verified Corpus Normalization Statistics (`preprocessed_data/*/stats.json`)

| Statistic | English (`Svarah/stats.json`) | Telugu (`IndicTTS/stats.json`) | Physical Units & Role |
|---|---|---|---|
| **Pitch Mean ($\mu_p$)** | `177.9316 Hz` | `195.6199 Hz` | Corpus $F_0$ mean for z-score & inverse normalization |
| **Pitch Std ($\sigma_p$)** | `59.4528 Hz` | `46.4196 Hz` | Corpus $F_0$ standard deviation |
| **Normalized Pitch Range $[p_{min}, p_{max}]$** | `[-2.9928, 10.1252]` | `[-2.9401, 11.8867]` | Bounds for 256 linear quantization bins (`pitch_bins`) |
| **Energy Mean ($\mu_e$)** | `27.3145` | `28.1071` | Corpus STFT $L_2$-norm mean |
| **Energy Std ($\sigma_e$)** | `30.3573` | `26.6611` | Corpus STFT $L_2$-norm standard deviation |
| **Normalized Energy Range $[e_{min}, e_{max}]$** | `[-0.8998, 7.4643]` | `[-1.0542, 8.6770]` | Bounds for 256 linear quantization bins (`energy_bins`) |

## 10. Model Configuration & Architectural Hyperparameters

Table 4 documents the exact model configurations from `config/Svarah/model.yaml` and `config/IndicTTS/model.yaml`. Both models contain approximately $35.1$M trainable parameters (differing only in the source vocabulary and speaker embedding table rows).

| Module | Hyperparameter | English (Svarah) | Telugu (IndicTTS) | Codebase Reference |
|---|---|---|---|---|
| **Text Encoder** | encoder_layer ($N_e$) | 4 | 4 | common/transformer/models.py |
| | encoder_head (h) / encoder_hidden (d) | 2 heads / 256 dim | 2 heads / 256 dim | ScaledDotProductAttention ($d_k = 128$) |
| | conv_filter_size / conv_kernel_size | 1024 / [9, 1] | 1024 / [9, 1] | PositionwiseFeedForward (Conv1D) |
| | encoder_dropout | 0.2 | 0.2 | Residual & attention dropout |
| **Variance Adaptor** | Predictor Conv1D Layers | 2 layers (256 ch) | 2 layers (256 ch) | common/model/variance_predictor.py |
| | kernel_size / dropout | 3 / 0.5 | 3 / 0.5 | Duration, Pitch & Energy predictors |
| | pitch.feature / energy.feature | frame_level | frame_level | Applied after LengthRegulator |
| | pitch_quantization / energy_quantization | linear (256 bins) | linear (256 bins) | torch.bucketize (255 boundaries) |
| **Mel Decoder** | decoder_layer ($N_d$) | 6 | 6 | common/transformer/models.py |
| | decoder_head (h) / decoder_hidden (d) | 2 heads / 256 dim | 2 heads / 256 dim | Matches encoder hidden dimension |
| | conv_filter_size / conv_kernel_size | 1024 / [9, 1] | 1024 / [9, 1] | PositionwiseFeedForward |
| | decoder_dropout | 0.2 | 0.2 | Residual & attention dropout |
| **PostNet** | Layers / Channels / Kernel Size | 5 / 512 / 5 | 5 / 512 / 5 | common/model/postnet.py (Mask-aware) |
| **Conditioning** | n_src_vocab (V) | 360 | 101 | nn.Embedding(V, 256, padding_idx=0) |
| | multi_speaker / n_speaker | True / 115 *(88 active)* | True / 2 | nn.Embedding(n_speaker, 256) |
| **Vocoder** | Architecture | HiFi-GAN Universal v1 | HiFi-GAN Universal v1 | original_fastspeech2/hifigan/ |

## 11. Training Setup & Optimization Regime

Both models were trained independently on NVIDIA T4/L4 GPUs (English/train.py, Telugu/train.py) under identical optimization hyperparameters (config/Svarah/train.yaml, config/IndicTTS/train.yaml):

- **Total Training Steps**: $20{,}000$ optimizer updates:
- **English (Svarah)**: $3{,}208$ train utterances at batch_size = 16 ($200$ steps/epoch) ⇒ **100 complete epochs** ($1$ h $23$ m $32$ s).
- **Telugu (IndicTTS)**: $2{,}300$ train utterances at batch_size = 16 ($\approx 143$ steps/epoch) ⇒ **140 complete epochs** ($1$ h $36$ m $45$ s).
- **Optimizer**: Adam ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon = 10^{-9}$, weight_decay = 0.0).
- **Noam Learning Rate Scheduler** (common/model/optimizer.py):

$$\text{lr}(\text{step}) = d_{\text{model}}^{-0.5} \cdot \min(\text{step}^{-0.5}, \text{step} \cdot \text{warmup_steps}^{-1.5}), \qquad d_{\text{model}} = 256, \quad \text{warmup_steps} = 4{,}000$$

The learning rate increases linearly during the first $4{,}000$ steps to a peak of $\text{lr}_{\max} = 256^{-0.5} \times 4000^{-0.5} = 9.882 \times 10^{-4}$, then decays proportionally to $\text{step}^{-0.5}$, reaching $4.419 \times 10^{-4}$ at step $20{,}000$.

- **Gradient Clipping**: Global $L_2$-norm threshold grad_clip_thresh $= 1.0$.
- **Logging & Validation Cadence**: Training metrics logged every 100 steps (201 records in training_history.csv); full validation sweep and checkpoint evaluation executed every 1,000 steps (20 validation evaluations in validation_history.csv).

## 12. Training and Validation Loss Analysis

### 12.1 Quantitative Validation Trajectories Across All Milestones

*Methodological Note: Absolute validation loss values cannot be compared directly across English and Telugu as a direct measure of model quality, because the two models operate over different token granularities (360 ARPAbet symbols vs. 101 Telugu graphemes), speaker cardinalities (88 vs. 2 speakers), and corpus normalization variances.*

| Global Step | En Epoch | En Val Total | En Mel $L_1$ | En Post $L_1$ | En Pitch | En Energy | En Dur | Te Epoch | Te Val Total | Te Mel $L_1$ | Te Post $L_1$ | Te Pitch | Te Energy | Te Dur |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1,000** | 5 | 2.8021 | 0.8243 | 0.9953 | 0.4524 | 0.2923 | 0.2378 | 7 | 2.5621 | 0.6853 | 0.7827 | 0.7292 | 0.2576 | 0.1072 |
| **3,000** | 15 | 2.4783 | 0.7219 | 0.7248 | 0.4648 | 0.2804 | 0.2864 | 21 | 2.2485 | 0.5562 | 0.5579 | 0.7932 | 0.2428 | 0.0984 |
| **5,000** | 25 | 2.4060 | 0.6934 | 0.6934 | 0.4747 | 0.2759 | 0.2686 | 35 | 2.1738 | 0.5248 | 0.5249 | 0.7940 | 0.2369 | 0.0932 |
| **8,000** | 40 | 2.3842 | 0.6812 | 0.6810 | 0.4669 | 0.2801 | 0.2750 | 56 | 2.1194 | 0.5108 | 0.5107 | 0.7685 | 0.2358 | 0.0936 |
| **10,000** | 50 | 2.3794 | 0.6767 | 0.6766 | 0.4670 | 0.2815 | 0.2775 | 70 | 2.1052 | 0.5072 | 0.5071 | 0.7630 | 0.2352 | 0.0927 |
| **13,000** | 65 | 2.3698 | 0.6741 | 0.6739 | 0.4638 | 0.2819 | 0.2761 | **91** | **2.0869** | **0.5039** | **0.5039** | **0.7481** | **0.2356** | **0.0954** |
| **15,000** | 75 | 2.3744 | 0.6775 | 0.6774 | 0.4656 | 0.2827 | 0.2712 | 105 | 2.1111 | 0.5069 | 0.5068 | 0.7648 | 0.2349 | 0.0978 |
| **19,000** | **95** | **2.3489** | **0.6686** | **0.6684** | **0.4592** | **0.2802** | **0.2725** | 133 | 2.1146 | 0.5065 | 0.5064 | 0.7671 | 0.2381 | 0.0965 |
| **20,000** | 100 | 2.3689 | 0.6746 | 0.6743 | 0.4596 | 0.2850 | 0.2754 | 140 | 2.1181 | 0.5070 | 0.5069 | 0.7688 | 0.2390 | 0.0964 |

## 12.2 Convergence Behaviour, Overfitting Analysis & Checkpoint Reconciliation

1. **Rapid Warmup Convergence (Steps 100–5,000)**:

During the 4,000-step Noam warmup phase, training total loss dropped steeply from $5.84 \rightarrow 1.52$ (English) and $4.91 \rightarrow 1.18$ (Telugu). Notably, at Step 1,000, PostNet $L_1$ loss was higher than Pre-Net Mel $L_1$ loss ($0.9953$ vs. $0.8243$ in English; $0.7827$ vs. $0.6853$ in Telugu) while the 5-layer convolutional residual network initialized its feature detectors; by Step 5,000, PostNet converged tightly onto the Mel $L_1$ trajectory.

2. **English Best Observed Checkpoint (Step 19,000, Validation Loss `2.34890`)**:

Because English conditions on 88 distinct voice profiles, acoustic reconstruction continued to refine steadily through late training, reaching its lowest observed validation total loss ($2.34890$), Mel $L_1$ loss ($0.6686$), and PostNet $L_1$ loss ($0.6684$) at **Step 19,000 (Cumulative Epoch 95)** before a minor stochastic fluctuation at Step 20,000 ($2.36890$). *(Provenance Note: Because the final Colab training session resumed from a saved checkpoint at Step 13,500 / Epoch 68, the internal checkpoint dictionary records session epoch `27` at Step 19,000, which corresponds to cumulative epoch $68 + 27 = 95$ in `validation_history.csv`).*

3. **Telugu Best Observed Checkpoint (Step 13,000, Validation Loss `2.08688`) & Post-13k Pitch Overfitting**:

Telugu validation loss reached its lowest observed value at **Step 13,000 (Epoch 91)**: Total $= 2.08688$, Mel $= 0.5039$, PostNet $= 0.5039$, Pitch $= 0.7481$. Between Steps 13,000 and 20,000, training loss continued to fall ($0.91 \rightarrow 0.7791$), but validation total loss rose to $2.11808$ driven primarily by **validation pitch MSE drifting upward from $0.7481$ to $0.7688$** as the 2-layer pitch predictor began memorizing training-utterance $F_0$ micro-contours. Selecting `checkpoint_best.pth.tar` at Step 13,000 prevented pitch over-smoothing. *(Provenance Note: At Step 20,000, the single-step validation sweep logged inside `checkpoint_latest.pth.tar` is `2.094272`, whereas the full Epoch 140 aggregate recorded in `validation_history.csv` is `2.118075`).*

## 12.3 Training & Validation Loss Curves

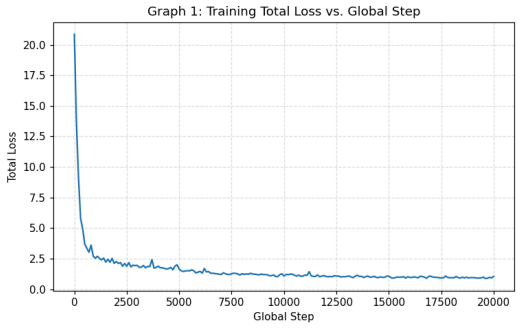



**Figure 3:** English Training Total Loss Curve (Steps 100–20,000)

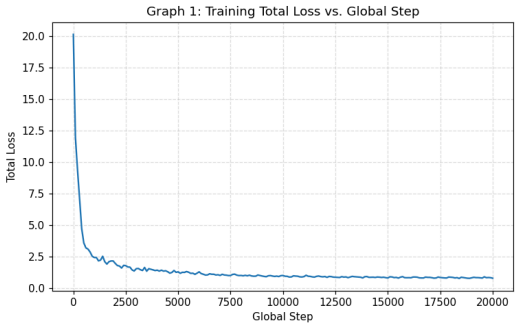



**Figure 4:** Telugu Training Total Loss Curve (Steps 100–20,000)

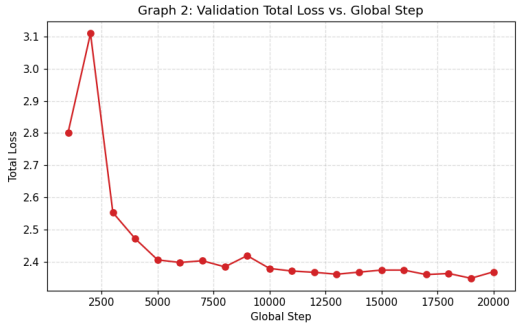



**Figure 5:** English Validation Total Loss Curve (Steps 1,000–20,000)

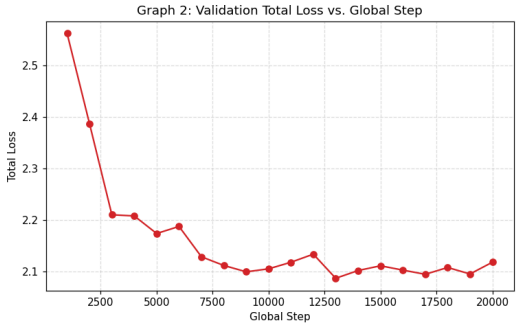



**Figure 6:** Telugu Validation Total Loss Curve (Steps 1,000–20,000)

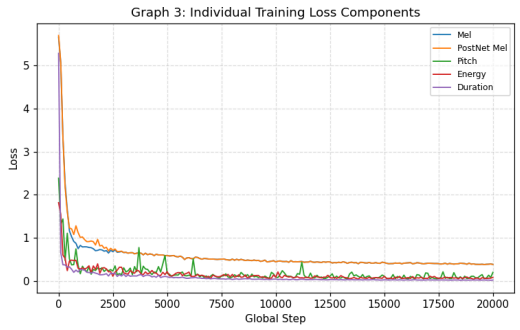



**Figure 7:** English Training Component Loss Breakdown

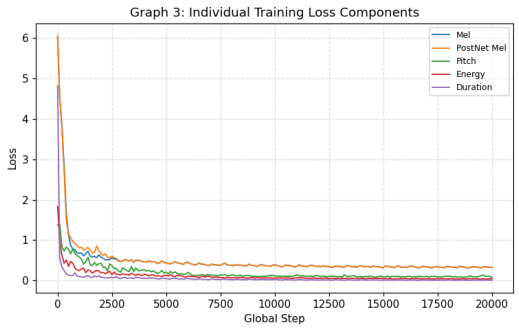



**Figure 8:** Telugu Training Component Loss Breakdown

## 13. Inference Methodology & Prosody Inverse-Normalization

During inference ( `English/inference.py` , `Telugu/inference.py` ), ground-truth targets ( `mels` , `p_targets` , `e_targets` , `d_targets` ) are set to `None` and prosody control multipliers default to $\alpha_d = \alpha_p = \alpha_e = 1.0$. To ensure scientific clarity across all tables and plots, we strictly distinguish six prosodic quantities:

1. **Normalized Pitch ($z_p$, dimensionless $z$-score)**: Raw frame-level output of `pitch_predictor` (plotted directly on English pitch plots).
2. **Inverse-Normalized Physical Pitch ($F_0$, in Hz)**: Reconstructed via corpus `stats.json` (plotted directly on Telugu pitch plots and tabulated for both languages):

$$F_{0,\,t}\,(\text{Hz}) = \hat{z}_{p,\,t} \cdot \sigma_p + \mu_p$$

3. **All-Frame Pitch vs. Voiced-Only Pitch**:

- **All-Frame Pitch**: Mean and standard deviation computed across all $L_{\text{mel}}$ frames, including silent `@sp` / `sp` pauses and unvoiced consonants.
- **Voiced-Only Pitch ($E_{\text{linear}} > 5.0$)**: Mean and standard deviation computed strictly over active voiced frames where physical STFT $L_2$-norm energy exceeds the silence floor ($E_{\text{linear},\,t} > 5.0$).

4. **Normalized Energy ($z_e$, dimensionless $z$-score)** vs. **Inverse-Normalized Physical Energy ($E_{\text{linear}}$, STFT $L_2$-norm)**:

$$E_{\text{linear},\,t} = \hat{z}_{e,\,t} \cdot \sigma_e + \mu_e, \qquad t_{\text{sec}} = \frac{\text{frame_idx} \times 256}{22{,}050}$$

## 14. Generated English Samples Analysis (E1, E2a, E2b, E3, E4)

All five English assignment sentences were synthesized using the best observed validation checkpoint ( `Step 19,000` , `checkpoint_best.pth.tar` ) conditioned on Speaker Profile `0` (Female Indian English) and vocoded via HiFi-GAN Universal at 22,050 Hz.

### 14.1 Master Quantitative Summary Table — English Test Set

| ID | Tokens ($L_{src}$) | Mel Frames ($L_{mel}$) | Audio Dur (s) | Token Dur Mean ± Std [Min, Max] | All-Frame Pitch $z_p$ | All-Frame Pitch $F_0$ (Hz) | Voiced Frames (E>5) | Voiced Pitch $F_0$ (Hz) | All-Frame Energy $z_e$ | Physical Energy $E_{linear}$ ($L_2$) |
|---|---|---|---|---|---|---|---|---|---|---|
| **E1** | 85 | 522 | 6.060 | 6.14 ± 2.05 [2, 11] | 1.111 ± 0.336 | 243.98 ± 19.98 | 505 / 522 | **243.93 ± 20.11** | 0.008 ± 0.444 | 27.54 ± 13.47 [2.38, 64.53] |
| **E2(a)** | 50 | 333 | 3.866 | 6.66 ± 2.40 [3, 12] | 0.903 ± 0.444 | 231.60 ± 26.42 | 312 / 333 | **230.10 ± 24.97** | -0.062 ± 0.468 | 25.44 ± 14.21 [2.86, 60.70] |
| **E2(b)** | 56 | 358 | 4.156 | 6.39 ± 2.61 [2, 12] | 0.882 ± 0.331 | 230.39 ± 19.69 | 331 / 358 | **230.36 ± 20.35** | -0.096 ± 0.473 | 24.39 ± 14.36 [2.02, 54.16] |
| **E3** | 37 | 240 | 2.786 | 6.49 ± 2.47 [3, 16] | 1.011 ± 0.321 | 238.05 ± 19.10 | 221 / 240 | **238.75 ± 19.44** | -0.021 ± 0.562 | 26.67 ± 17.05 [2.77, 74.45] |
| **E4** | 81 | 567 | 6.583 | 7.00 ± 4.75 [2, 40] | 1.016 ± 0.300 | 238.35 ± 17.85 | 489 / 567 | **237.70 ± 18.61** | -0.051 ± 0.566 | 25.78 ± 17.17 [0.05, 80.48] |

### 14.2 Individual Sample Evaluations (E1 through E4)

**Sample E1 — Polysyllabic Morphological Stress Test**

- **Exact Input Text**: *"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."*
- **Phonetic Transcription (85 ARPAbet tokens)**: The 28-letter word *"antidisestablishmentarianism"* was decomposed via `g2p_en` into 28 phonemes ( `AE2 N T AY0 D IH0 S AH0 S T AE2 B L IH0 SH M AH0 N T EH1 R IY0 AH0 N IH0 Z AH0 M` , tokens 2–29) carrying secondary stress on `AE2` and primary stress on `EH1` .
- **Duration Analysis**: Total duration is 522 frames (6.060 s), with a compact mean of 6.14 ± 2.05 frames/token. Within *"antidisestablishmentarianism"*, the model allocated 166 frames (1.93 s, averaging 5.93 frames/phoneme), peaking at 10 frames (116 ms) on the palato-alveolar fricative `SH` (token 16) and 8 frames on stressed diphthongs/vowels ( `AY0` , `IY0` ). Utterance-final lengthening occurs on `"proceedings"` ( `IH0 NG Z` = `[5, 10, 11]` frames).
- **Pitch & Energy Analysis**: Voiced $F_0$ averages 243.93 ± 20.11 Hz ($z_p$ = 1.110 ± 0.338). Pitch reaches its global peak early on the compound word at frame 53 (0.62 s, 298.8 Hz, $z_p$ = 2.033) and declines smoothly to 191.4 Hz ($z_p$ = 0.227) at frame 521. Physical energy averages 27.54 ± 13.47, peaking at 64.53 ($z_e$ = 1.226) at frame 392 (4.55 s) on the stressed syllable `AA2 R` of *"parliamentary"*.
- **Pronunciation & Speech Quality**: Highly intelligible, artifact-free articulation of all 12 syllables of *"antidisestablishmentarianism"*. Because the Length Regulator enforces deterministic frame expansion, the model completely avoids the attention-skipping and syllable-looping failures typical of autoregressive TTS on long morphological chains.

**Sample E2(a) — Contextual Homograph "conduct" (Noun Role)**

- **Exact Input Text**: *"The orchestra's conduct during the ceremony was praised by the audience."*
- **Phonetic Transcription (50 ARPAbet tokens)**: `"conduct"` (tokens 11–17) is transcribed via CMUdict as `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']` .
- **Duration Analysis**: Total duration is 333 frames (3.866 s), averaging 6.66 ± 2.40 frames/token. The noun *"conduct"* spans 42 frames (0.488 s) with per-phoneme durations `[6, 4, 10, 4, 5, 8, 5]` (first syllable `K AA1 N` = 20 frames; second syllable `D AH0 K T` = 22 frames). Longest tokens are the sibilants `S` in *"ceremony"* (token 24, 12 frames) and utterance-final `S` in *"audience"* (token 49, 12 frames).
- **Pitch & Energy Analysis**: Voiced $F_0$ averages 230.10 ± 24.97 Hz ($z_p$ = 0.877 ± 0.420). The global pitch maximum occurs directly on the primary-stressed first syllable `AA1 N` of *"conduct"* at frame 70 (0.81 s, 289.7 Hz, $z_p$ = 1.881), followed by a terminal declarative fall to 146.6 Hz ($z_p$ = -0.527) at frame 332. Physical energy averages 25.44 ± 14.21, peaking at 60.70 ($z_e$ = 1.100) at frame 14 on `"orchestra's"` ( `AO1 R` ).
- **Pronunciation & Speech Quality**: Natural, clear Indian English pronunciation. First-syllable primary stress ( `AA1` ) combined with the 289.7 Hz pitch peak yields an authentic noun realization (/ˈkɑːndʌkt/) matching the syntactic context.

**Sample E2(b) — Contextual Homograph "conduct" (Verb Role)**

- **Exact Input Text**: *"The researchers will conduct an experiment to evaluate the proposed method."*
- **Phonetic Transcription (56 ARPAbet tokens)**: Because CMUdict is a static dictionary without POS disambiguation, `"conduct"` (tokens 12–18) receives the **identical** phoneme sequence `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']` as in E2(a).
- **Duration Analysis**: Total duration is 358 frames (4.156 s), averaging 6.39 ± 2.61 frames/token. Here `"conduct"` spans 46 frames (0.534 s) with durations `[10, 5, 8, 3, 7, 8, 5]` (+4 frames longer than E2a due to pre-boundary lengthening on initial `K` [10 frames] after auxiliary `"will"` ). Longest utterance tokens are `CH` (12 frames) and `ER0 Z` ([11, 11] frames) in *"researchers"* and `OW1` (11 frames) in *"proposed"*.
- **Pitch & Energy Analysis**: Voiced $F_0$ averages 230.36 ± 20.35 Hz ($z_p$ = 0.882 ± 0.342), with a narrower pitch range (179.8 – 272.9 Hz) than E2(a). Over `"conduct"` , mean pitch is lower ($z_p$ = 1.022, peak 1.423 vs. 1.881 in E2a). Physical energy averages 24.39 ± 14.36, peaking at 54.16 ($z_e$ = 0.884) at frame 50 on *"researchers"*.
- **Pronunciation & Speech Quality**: Segmentally crisp and fluent across all words (*"experiment"*, *"evaluate"*, *"proposed"*), but *"conduct"* retains noun-like first-syllable vowel quality ( `AA1` ) rather than shifting to verb-stressed /kənˈdʌkt/ because of the static lexicon lookup.

**Sample E3 — Orthographic Split Anomaly ("rhyth m")**

- **Exact Input Text**: *"The rhyth m of the orchestra changed unexpectedly."*
- **Phonetic Transcription (37 ARPAbet tokens)**: Preserving the exact prompt text containing a space inside `"rhyth m"` caused the tokenizer to process `"rhyth"` (OOV → `g2p_en` : `R AY1 DH` , tokens 2–4) and `"m"` (letter-name lookup: `EH1 M` , tokens 5–6) as two distinct words rather than canonical `R IH1`

`DH AH0 M`.

- **Duration Analysis**: Shortest English utterance at 240 frames (2.786 s), averaging $6.49 \pm 2.47$ frames/token. The fragmented sequence `R AY1 DH EH1 M` consumes $4 + 10 + 7 + 6 + 7 = 34$ frames (0.395 s), with 20 frames (232 ms) across the `DH EH1 M` juncture. Utterance-final `IY0` in *"unexpectedly"* undergoes strong phrase-final lengthening (16 frames / 186 ms).
- **Pitch & Energy Analysis**: Voiced $F_0$ averages $238.75 \pm 19.44$ Hz ($z_p = 1.023 \pm 0.327$), peaking at frame 116 (1.35 s, 273.7 Hz, $z_p = 1.611$) on `"changed"` ( `CH EY1 N JH D` ) and reaching its minimum (192.9 Hz, $z_p = 0.251$) at frame 213. Physical energy averages $26.67 \pm 17.05$ with a sharp peak of 74.45 ($z_e = 1.553$) at frame 32 (0.37 s) on the open vowel `EH1` of `"m"`.
- **Pronunciation & Speech Quality**: Directly demonstrates how non-autoregressive TTS faithfully renders front-end tokenization: the split `"rhyth m"` is audibly pronounced as /raɪð ɛm/, whereas the remainder of the sentence (*"of the orchestra changed unexpectedly"*) is synthesized with smooth, natural prosody.

**Sample E4 — Temporal Numerals, Dates & Code-Mixed OOV ("andaroo")**

- **Exact Input Text**: *"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."*
- **Phonetic Transcription (81 ARPAbet tokens)**: `English/text/normalizer.py` expanded `"10:45 AM"` and `"12 November 2026"` into words, while the code-mixed Telugu-English word `"andaroo"` (meaning *"everyone"*) was handled by `g2p_en` as `['AE0', 'N', 'D', 'AA1', 'R', 'OW0']` (tokens 64–69). Two explicit `sp` pause tokens were inserted at the colon (token 20, 7 frames) and comma (token 61, 40 frames / 0.464 s).
- **Duration Analysis**: Longest English utterance at 567 frames (6.583 s), averaging $7.00 \pm 4.75$ frames/token. The comma pause `sp` at token 61 receives 40 frames (464 ms), cleanly separating the temporal clause from the concluding clause *"so andaroo should arrive on time"*.
- **Pitch & Energy Analysis**: Voiced $F_0$ averages $237.70 \pm 18.61$ Hz ($z_p = 1.005 \pm 0.313$, across 489 voiced frames), peaking at 292.0 Hz ($z_p = 1.918$) at frame 29 ( `"meeting"` ). During the 40-frame comma pause (frames 403–442), physical energy drops to near-zero (0.05 $L_2$, $z_e = -0.898$) before recovering cleanly on *"so andaroo"* (46.2 $L_2$). Global energy peaks at 80.48 ($z_e = 1.751$) at frame 178 (2.07 s).
- **Pronunciation & Speech Quality**: Clear clause boundary pausing and smooth code-mixed rendering of *"andaroo"* (/ændɑːroʊ/). Minor regex interaction in the colon/hyphen normalizer contracted the middle syllables of `"forty-five"` and `"six"` in `"twenty-six"`, which is documented as a text-normalization edge case in Section 18.

## 15. Generated Telugu Samples Analysis (T1, T2a, T2b, T3, T4)

All five Telugu assignment sentences were synthesized using the best observed validation checkpoint ( `Step 13,000` , `checkpoint_best.pth.tar` ) conditioned on Speaker `spk_001` (Male Native Telugu) in `telugu_digits` mode using the verified text manifest `Results/Telugu/official_samples.json` .

### 15.1 Master Quantitative Summary Table — Telugu Test Set

| ID | Tokens ($L_{src}$) | `@sp` Pauses (Frames) | Mel Frames ($L_{mel}$) | Audio Dur (s) | Token Dur Mean ± Std [Min, Max] | All-Frame Pitch $z_p$ | All-Frame Pitch $F_0$ (Hz) | Voiced Frames (E>5) | Voiced Pitch $F_0$ (Hz) | All-Frame Energy $z_e$ | Physical Energy $E_{linear}$ ($L_2$) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **T1** | 67 | 5 (68f) | 464 | 5.387 | 6.93 ± 4.32 [1, 18] | -0.073 ± 0.968 | 192.25 ± 44.93 | 377 / 464 | **197.25 ± 39.30** | 0.289 ± 1.102 | 35.82 ± 29.37 [-0.22, 116.54] |
| **T2(a)** | 33 | 5 (74f) | 283 | 3.286 | 8.58 ± 5.57 [1, 20] | -0.118 ± 0.887 | 190.15 ± 41.19 | 206 / 283 | **196.00 ± 37.46** | 0.160 ± 1.081 | 32.37 ± 28.82 [-0.25, 114.51] |
| **T2(b)** | 39 | 5 (91f) | 291 | 3.378 | 7.46 ± 5.77 [1, 26] | -0.278 ± 0.953 | 182.69 ± 44.22 | 201 / 291 | **186.67 ± 44.03** | -0.126 ± 0.877 | 24.76 ± 23.37 [-0.20, 83.00] |
| **T3** | 32 | 3 (37f) | 208 | 2.415 | 6.50 ± 4.68 [1, 17] | -0.475 ± 0.783 | 173.55 ± 36.32 | 165 / 208 | **179.57 ± 31.53** | -0.086 ± 0.772 | 25.81 ± 20.57 [0.01, 102.67] |
| **T4** | 89 | 13 (144f) | 692 | 8.034 | 7.78 ± 4.55 [1, 22] | -0.089 ± 0.806 | 191.50 ± 37.40 | 541 / 692 | **195.47 ± 33.15** | 0.168 ± 0.946 | 32.59 ± 25.23 [-0.32, 106.46] |

### 15.2 Individual Sample Evaluations (T1 through T4)

**Sample T1 — Polysyllabic Technical Compound & Complex Conjuncts**

- **Exact Input Text**: న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
- **Grapheme Tokenization (67 tokens, 0 `<unk>` )**: The 20-grapheme transliterated compound `న్యూరోఫార్మకోలాజికల్` (*"neuropharmacological"*, tokens 0–19: `['న', '్', 'యూ', 'ర', 'ో', 'ఫ్', 'ా', 'ర్', '్', 'మ', 'క', 'ో', 'ల', 'ా', 'జ', 'ి', 'క్', 'ల్', '్']` ) is tokenized as a single uninterrupted lexical unit followed by 5 inter-word `@sp` pauses ( `[6, 18, 13, 13, 18]` frames; 68 pause frames total).
- **Duration Analysis**: Total duration is 464 frames (5.387 s), averaging 6.93 ± 4.32 frames/token. The 20-token compound `న్యూరోఫార్మకోలాజికల్` spans 120 frames (1.393 s), with virama ( `్` ) closures taking 1-2 frames and long vowel matras ( `ూ` , `ో` , `ా` ) taking 5-7 frames. Word-final vowels before `@sp` pauses exhibit strong pre-boundary lengthening ( `ో` in `పరిశోధనలో` = 18 frames; `ి` in `ప్రభావాన్ని` = 18 frames).
- **Pitch & Energy Analysis**: All-frame pitch is 192.25 ± 44.93 Hz ($z_p$ = -0.073 ± 0.968); voiced-only pitch (377 active frames) is 197.25 ± 39.30 Hz. Pitch peaks early on `న్యూరోఫార్మకోలాజికల్` at frame 22 (0.26 s, 281.5 Hz, $z_p$ = 1.851) and dips to 87.5 – 115 Hz during the 18-frame `@sp` pauses. Physical energy averages 35.82 ± 29.37, peaking at 116.54 ($z_e$ = 3.317) at frame 343 (3.98 s) on the open `ా` vowel of `ప్రభావాన్ని` .
- **Pronunciation & Speech Quality**: AI-assisted subjective rating: Clarity 4.0/5, Fluency 3.5/5, Naturalness 3.0/5 (Mean 3.50/5). All complex conjuncts ( `న్యూ` , `ర్మ` , `ప్ర` , `న్ని` , `శ్రే` ) are articulated accurately without phoneme dropping, though the 20-grapheme English loanword exhibits a slightly mechanical syllable-timed cadence across frames 1–120.

**Sample T2(a) — Contextual Homograph "పట్టు" (Physical Grip / Hold Meaning)**

- **Exact Input Text**: తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
- **Grapheme Tokenization (33 tokens, 0 `<unk>` )**: Contains 5 inter-word `@sp` tokens ( `[13, 20, 11, 20, 10]` frames; 74 pause frames total). The target word `పట్టు` (*"grip/hold"*, tokens 12–16) is tokenized as `['ప', 'ట', '్', 'ట', 'ు']` .
- **Duration Analysis**: Total duration is 283 frames (3.286 s), averaging 8.58 ± 5.57 frames/token. The target word `పట్టు` receives **exactly 20 frames (0.232 s)** distributed as `[5, 4, 2, 7, 2]` : 5 frames for onset `ప` , 4 + 2 + 7 = 13 frames for the geminate retroflex stop closure and release `ట్టు` ( `ట + ్ + ట` ), and 2 frames for short vowel `ు` . Diphthong `ై` in `తాడుపై` (18 frames) and long `ా` in `బలంగా` (18 frames) receive prominent vowel durations.
- **Pitch & Energy Analysis**: All-frame pitch is 190.15 ± 41.19 Hz; voiced-only pitch (206 frames) is 196.00 ± 37.46 Hz. Across `పట్టు` (frames 111–130), mean pitch is 199.2 Hz (range 169.4 – 249.7 Hz). Utterance pitch peaks at frame 99 (1.15 s, 329.3 Hz) at the onset transition following `@sp` . Physical energy averages 32.37 ± 28.82, peaking at 114.51 ($z_e$ = 3.241) at frame 30 ( `తాడుపై` ).
- **Pronunciation & Speech Quality**: Highest-rated Telugu sample (Clarity 4.5/5, Fluency 4.0/5, Naturalness 3.5/5, Mean 4.00/5). Crisp retroflex gemination on `పట్టు` and clear nasalization ( `ం` ) on `బలంగా ఉంది` .

**Sample T2(b) — Contextual Homograph "పట్టు" (Silk Fabric Meaning)**

- **Exact Input Text**: ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
- **Grapheme Tokenization (39 tokens, 0 `<unk>` )**: Contains 5 inter-word `@sp` tokens ( `[26, 20, 17, 11, 17]` frames; 91 pause frames total). The target word `పట్టు` (*"silk"*, tokens 4–8) is tokenized as the **identical** sequence `['ప', 'ట', '్', 'ట', 'ు']` .
- **Duration Analysis**: Total duration is 291 frames (3.378 s), averaging 7.46 ± 5.77 frames/token. Remarkably, `పట్టు` in T2(b) also receives **identically 20 frames (0.232 s)** with sub-token durations `[5, 5, 1, 7, 2]` (differing by just ± 1 frame between implosion `ట` and virama `్` compared to `[5, 4, 2, 7, 2]` in T2a). The utterance-final verb `వచ్చింది` ( `['వ', 'చ', '్', 'చ', 'ి', 'ం', 'ది', 'ీ']` , tokens 31–38) is compressed into 28 frames (0.325 s), with the geminate palatal affricate `చ్చ` ( `చ + ్ + చ` ) allocated only 3 + 2 + 1 = 6 frames.
- **Pitch & Energy Analysis**: All-frame pitch is 182.69 ± 44.22 Hz; voiced-only pitch (201 frames) is 186.67 ± 44.03 Hz. Over `పట్టు` (frames 62–81), mean pitch is 206.7 Hz (+7.5 Hz higher than T2a due to its earlier sentence position as modifier of `చీరను` ). Physical energy is lower overall (24.76 ± 23.37, peak 83.00 at frame 158 on `ధరించి` ), with reduced coda energy on `వచ్చింది` .
- **Pronunciation & Speech Quality**: AI-assisted rating: Clarity 2.5/5, Fluency 2.5/5, Naturalness 2.0/5 (Mean 2.33/5). While `ఆమె పట్టు చీరను` is clear, rapid duration allocation and lower energy on `ధరించి` and `వచ్చింది` cause mild perceptual softening of the palatal affricates.

**Sample T3 — Short Declarative Baseline & Sibilant Geminate ( `నిశ్శబ్దంగా` )**

- **Exact Input Text**: పులి అడవిలో నిశ్శబ్దంగా నడిచింది.

- **Grapheme Tokenization (32 tokens, 0 `<unk>`)**: Shortest Telugu utterance with 3 `@sp` pauses ( `[8, 16, 13]` frames; 37 pause frames total). Contains the complex double-conjunct word `నిశ్శబ్దంగా` ( `['న', 'ి', 'శ', '్', 'శ', 'బ', '్', 'ద', 'ం', 'గ', 'ా']` , tokens 12–22), which combines a geminate palatal sibilant ( `శ్శ` ) and a voiced labial-dental stop cluster ( `బ్ద` ).
- **Duration Analysis**: Total duration is 208 frames (2.415 s), averaging 6.50 ± 4.68 frames/token. `నిశ్శబ్దంగా` spans 79 frames (0.917 s), with `శ్శ` receiving 8 + 1 + 7 = 16 frames, `బ్ద` receiving 6 + 2 + 5 = 13 frames, anusvara `ం` receiving 12 frames, and final `గా` receiving 1 + 16 = 17 frames.
- **Pitch & Energy Analysis**: Exhibits the most compact pitch contour in the Telugu test set: all-frame pitch is 173.55 ± 36.32 Hz and voiced-only pitch (165 frames) is 179.57 ± 31.53 Hz (peaking at 275.5 Hz at frame 160). Physical energy averages 25.81 ± 20.57, peaking at 102.67 ($z_e = 2.797$) at frame 44 on `అడవిలో`, but tapering to < 8.0 on the final syllable `ది` of `నడిచింది` .
- **Pronunciation & Speech Quality**: AI-assisted rating: Clarity 3.5/5, Fluency 3.0/5, Naturalness 2.5/5 (Mean 3.00/5). Natural rhythm and accurate realization of `నిశ్శబ్దంగా` , with slight perceptual attenuation on the utterance-final syllable `ది` .

**Sample T4 — Long Compound Utterance with Native Telugu Digits**

- **Exact Input Text**: ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.
- **Grapheme Tokenization (89 tokens, 0 `<unk>`)**: Longest test utterance across both languages (89 tokens, 13 `@sp` pauses totaling 144 pause frames). In `telugu_digits` mode, ASCII numerals `2026` , `12` , and `10:45` map to native Telugu digit tokens `['౨', '౦', '౨', '౬']` (tokens 10–13), `['౧', '౨']` (tokens 22–23), and `['౧', '౦', '@sp', '౪', '౫']` (tokens 36–40).
- **Duration Analysis**: Spans 692 mel frames (8.034 s), averaging 7.78 ± 4.55 frames/token. However, each Telugu digit character ( `౨` , `౦` , `౬` , `౧` , `౪` , `౫` ) is a single token in the vocabulary and receives only 3 − 11 frames (35 − 128 ms) per digit: `౨౦౨౬` ( `2026` ) is compressed into 11 + 8 + 7 + 9 = 35 frames (0.406 s) and `౧౦ @sp ౪౫` ( `10:45` ) into 7 + 3 + 5 + 6 + 9 = 30 frames (0.348 s), whereas spoken Telugu words for those numbers require 12–18 graphemes. Major clausal boundaries at `తేదీ` ( `@sp` token 30 = 22 frames) and the comma after `ప్రారంభమవుతుంది` ( `@sp` token 64 = 21 frames) provide clear phrasal segmentation.
- **Pitch & Energy Analysis**: All-frame pitch is 191.50 ± 37.40 Hz; voiced-only pitch (541 frames) is 195.47 ± 33.15 Hz (peaking at 331.3 Hz at frame 657). Physical energy averages 32.59 ± 25.23, peaking at 106.46 ($z_e = 2.939$) at frame 169 ( `నవంబర్` ).
- **Pronunciation & Speech Quality**: AI-assisted rating: Clarity 2.0/5, Fluency 2.0/5, Naturalness 1.5/5 (Mean 1.83/5). While pure lexical words ( `సమావేశం` , `ప్రారంభమవుతుంది` , `సమయానికి` ) are well-formed, single-token digit compression on `2026` , `12వ` , and `10:45` causes perceptual rushing, motivating our `--digit_mode words` lexical expansion recommendation in Section 19.

## 16. Variance Adaptor Plot Analysis (All 30 Contours)

Below is the complete visual and quantitative inspection of all **30 Variance Adaptor plots** ( `10 Duration` , `10 Pitch` , and `10 Energy` contours) generated from the ten canonical assignment sentences.

### 16.1 English Sample E1 — Variance Adaptor Contours

- **Duration Plot ( `E1_duration.png` )**: Across 85 ARPAbet tokens (522 frames), durations remain tightly bounded between 2 and 11 frames ($\mu = 6.14$, $\sigma = 2.05$). Stressed vowels ( `AY0` [8f], `IY0` [8–10f], `EY1` [10f]) and fricatives/nasals ( `SH` [10f], `M` [10f], `S` [10f], final `Z` [11f]) form clear local peaks, while unstressed schwas ( `AH0` [2–4f]) and stop closures ( `T` , `D` , `B` [3–5f]) form troughs.
- **Pitch Plot ( `E1_pitch.png` )**: Normalized pitch $z_p$ begins high on *"The antidisestablishmentarianism"* (peaking at $z_p = 2.033$ / 298.8 Hz at frame 53), resets to local accent peaks on *"movement gained"* ($z_p = 1.48$), *"momentum"* ($z_p = 1.45$), and *"parliamentary"* ($z_p = 1.38$), and exhibits classic declarative declination down to $z_p = 0.227$ (191.4 Hz) at frame 521.
- **Energy Plot ( `E1_energy.png` )**: Normalized energy $z_e$ ($\mu = 0.008$, $\sigma = 0.444$) oscillates rhythmically at syllable rate, rising to +0.8 to +1.23 on vocalic nuclei (global peak $z_e = 1.226$ / 64.53 $L_2$ at frame 392 on *"parliamentary"*) and dipping to −0.75 to −0.82 during voiceless stop closures ( `T` , `P` , `K` ).

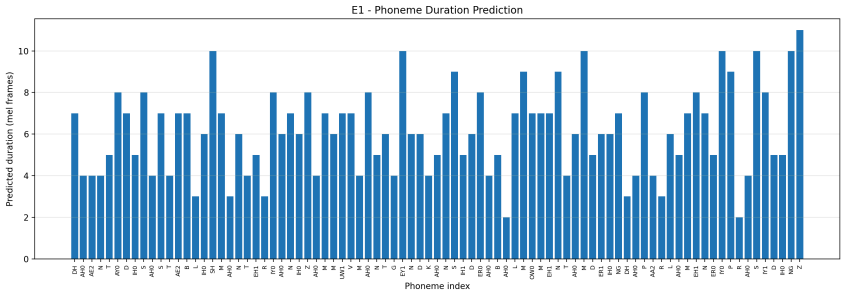



**Figure 9:** Sample E1 Duration Contour

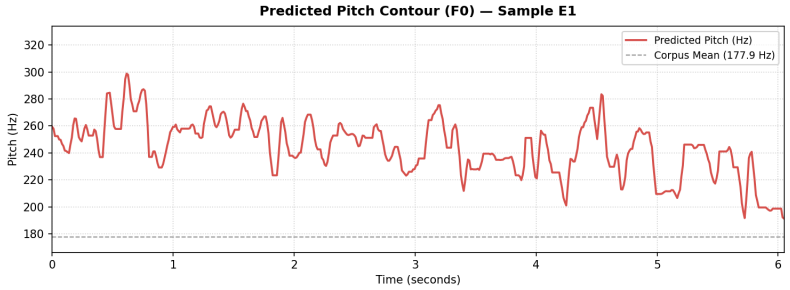



**Figure 10:** Sample E1 Pitch Contour

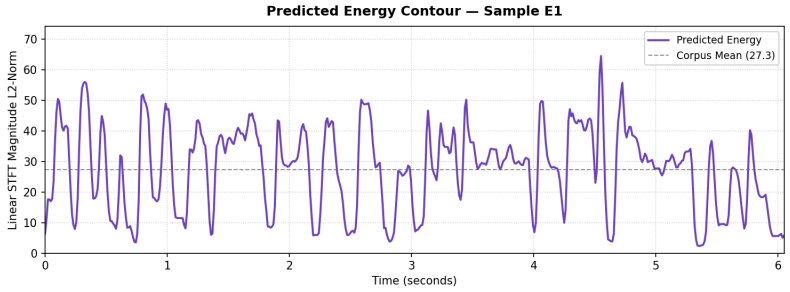



**Figure 11:** Sample E1 Energy Contour

### 16.2 English Sample E2(a) — Variance Adaptor Contours ("conduct" as Noun)

- **Duration Plot (`E2a_duration.png`)**: Across 50 tokens (333 frames, $\mu = 6.66 \pm 2.40$), sibilant fricatives (`S` in *"ceremony"* [12f], final `S` in *"audience"* [12f], `Z` in *"orchestra's"* [11f]) and nasal codas (`N` in *"conduct"* [10f] and *"audience"* [10f]) dominate token durations. The noun *"conduct"* (`K AA1 N D AH0 K T`, tokens 11–17) spans 42 frames (`[6, 4, 10, 4, 5, 8, 5]`).
- **Pitch Plot (`E2a_pitch.png`)**: Exhibits a pronounced focal pitch accent on *"conduct"*, rising sharply to the utterance maximum $z_p = 1.881$ (289.7 Hz) at frame 70 on the primary-stressed first syllable `AA1 N`, followed by secondary peaks on *"ceremony"* ($z_p = 1.34$) and *"praised"* ($z_p = 1.29$), and a steep terminal fall to $z_p = -0.527$ (146.6 Hz) at frame 332.
- **Energy Plot (`E2a_energy.png`)**: Energy peaks strongly on the initial stressed vowel `AO1` of *"orchestra's"* ($z_e = 1.100$ / 60.70 $L_2$ at frame 14) and on *"ceremony"* ($z_e = 0.89$) and *"audience"* ($z_e = 0.94$), with sharp $-0.80$ troughs marking plosive closures (`K`, `T`, `P`).

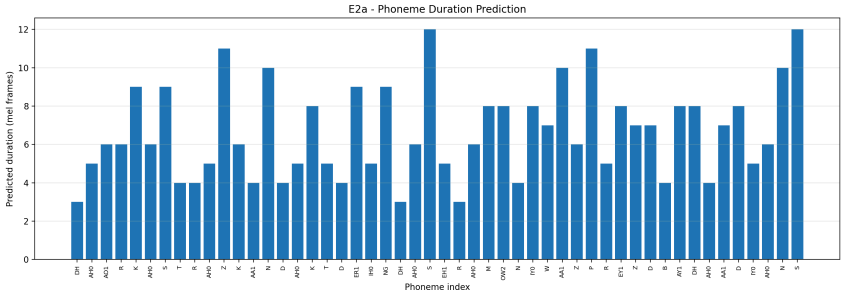



**Figure 12:** Sample E2a Duration Contour

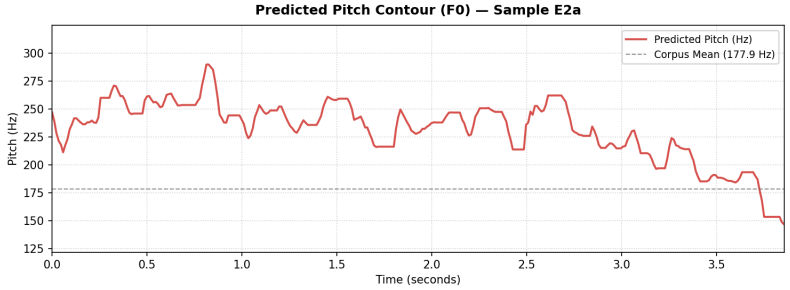



**Figure 13:** Sample E2a Pitch Contour

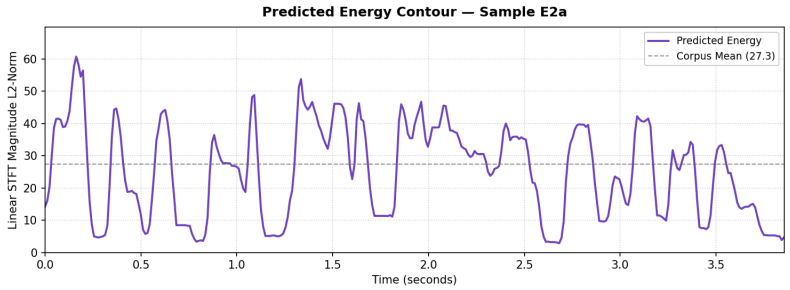



**Figure 14:** Sample E2a Energy Contour

### 16.3 English Sample E2(b) — Variance Adaptor Contours ("conduct" as Verb)

- **Duration Plot ( `E2b_duration.png` )**: Across 56 tokens (358 frames, $\mu = 6.39 \pm 2.61$), prominent durations appear on *"researchers"* ( `CH ER0 Z` = `[12, 11, 11]` frames), *"proposed"* ( `OW1 Z` = `[11, 9]` frames), and final `D` in *"method"* (10 frames). Verb *"conduct"* (tokens 12–18) spans 46 frames ( `[10, 5, 8, 3, 7, 8, 5]` ), with initial `K` lengthened to 10 frames after `"will"`.
- **Pitch Plot ( `E2b_pitch.png` )**: Unlike E2(a), pitch in E2(b) starts at $z_p = 1.597$ (272.9 Hz) on `"The researchers"` and remains flatter across the verb phrase `"will conduct"` (local peak $z_p = 1.423$, mean 1.022), reflecting reduced prosodic prominence in predicate verb position before declining to $z_p = 0.031$ (179.8 Hz) at frame 357.
- **Energy Plot ( `E2b_energy.png` )**: Energy is slightly more uniform ($\mu = -0.096 \pm 0.473$, max $z_e = 0.884$ / 54.16 $L_2$ at frame 50 on *"researchers"*), with 8 distinct syllabic peaks corresponding to the 8 lexical words of the sentence.

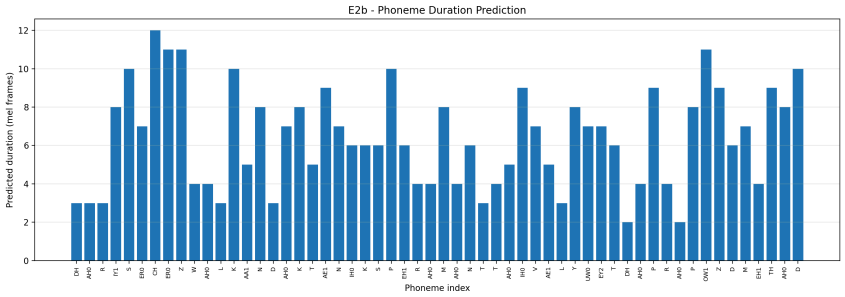



**Figure 15:** Sample E2b Duration Contour

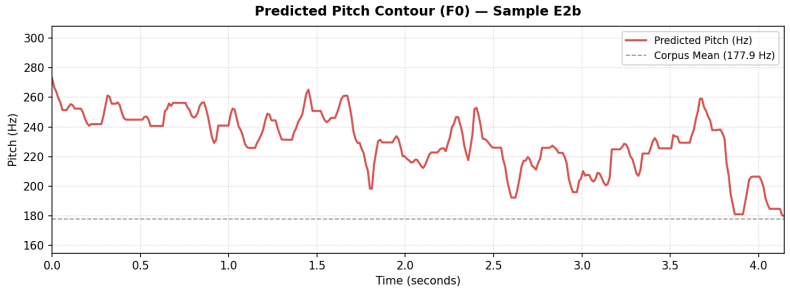



**Figure 16:** Sample E2b Pitch Contour

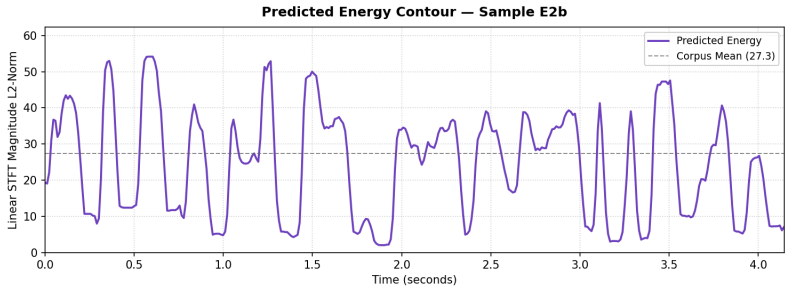



**Figure 17:** Sample E2b Energy Contour

### 16.4 English Sample E3 — Variance Adaptor Contours ("rhyth m" Split)

- **Duration Plot ( `E3_duration.png` )**: Across 37 tokens (240 frames, $\mu = 6.49 \pm 2.47$), the split tokens `R AY1 DH` ( `[4, 10, 7]` ) and `EH1 M` ( `[6, 7]` ) occupy 34 frames at the start of the plot, while utterance-final `IY0` in *"unexpectedly"* reaches the maximum duration of 16 frames (186 ms).
- **Pitch Plot ( `E3_pitch.png` )**: Shows a double-peaked contour across the separated fragments `"rhyth"` ($z_p = 1.36$) and `"m"` ($z_p = 1.28$), a mid-sentence focal peak on `"changed"` at frame 116 ($z_p = 1.611$ / 273.7 Hz), and a trough of $z_p = 0.251$ (192.9 Hz) at frame 213 on `"unexpectedly"`.
- **Energy Plot ( `E3_energy.png` )**: Displays a prominent spike at frame 32 ($z_e = 1.553$ / 74.45 $L_2$) directly on the isolated vowel `EH1` of `"m"`, visually confirming the acoustic salience of the orthographic split.

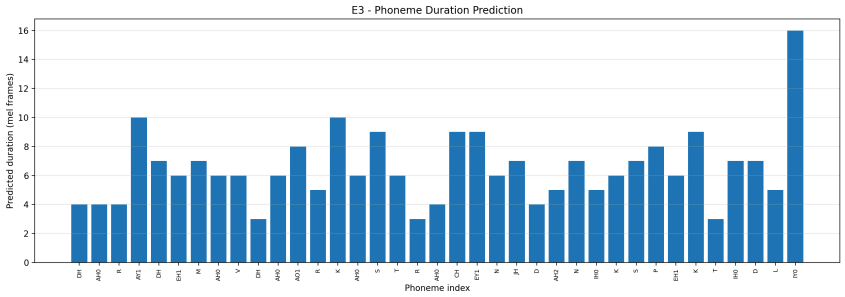



**Figure 18:** Sample E3 Duration Contour

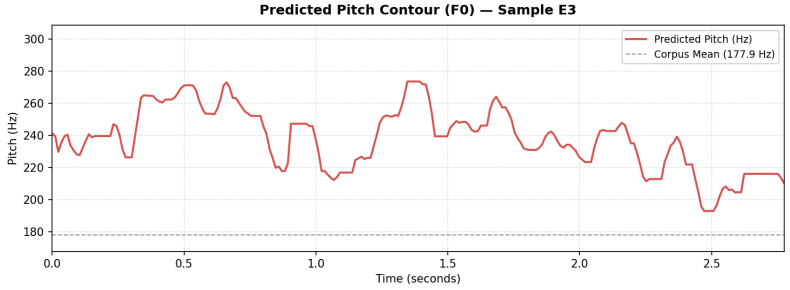



**Figure 19:** Sample E3 Pitch Contour

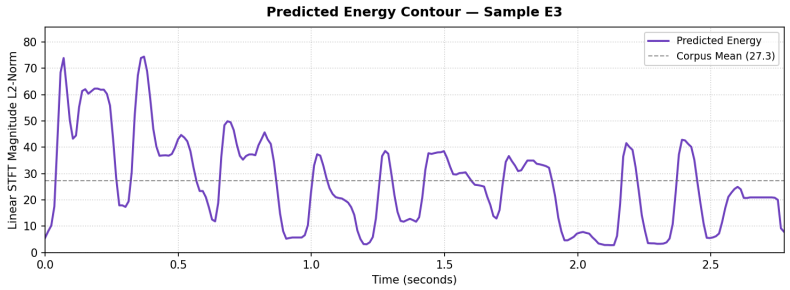



**Figure 20:** Sample E3 Energy Contour

### 16.5 English Sample E4 — Variance Adaptor Contours (Numbers, Dates & "andaroo")

- **Duration Plot ( `E4_duration.png` )**: Across 81 tokens (567 frames, $\mu = 7.00 \pm 4.75$), the standout feature is the 40-frame (464 ms) comma pause `sp` at token 61, alongside stressed diphthong `AY1` (18 frames at token 59; 12 frames at token 74 in *"arrive"*).
- **Pitch Plot ( `E4_pitch.png` )**: Starts with a high peak on *"meeting"* ($z_p = 1.918$ / 292.0 Hz at frame 29), maintains a steady plateau ($z_p \approx 0.8 - 1.4$) across the date/time expression, drops during the 40-frame comma pause (frames 403–442), and executes a clear pitch reset ($z_p = 1.52$) at the onset of the second clause *"so andaroo should arrive on time"*.
- **Energy Plot ( `E4_energy.png` )**: Reaches its global maximum at frame 178 ($z_e = 1.751$ / 80.48 $L_2$) on the time expression, drops flat to the floor ($z_e = -0.898$ / 0.05 $L_2$) across the 40-frame comma pause around $4.7 - 5.1$ s, and resumes rhythmic syllabic pulses through 6.58 s.

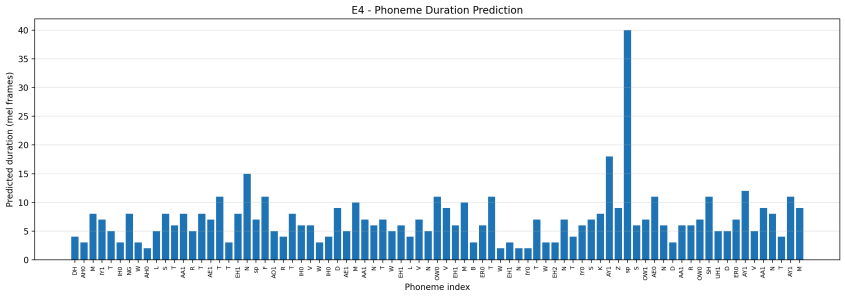



**Figure 21:** Sample E4 Duration Contour

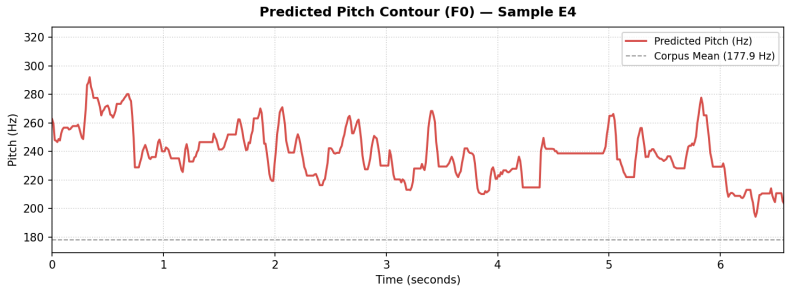



**Figure 22:** Sample E4 Pitch Contour

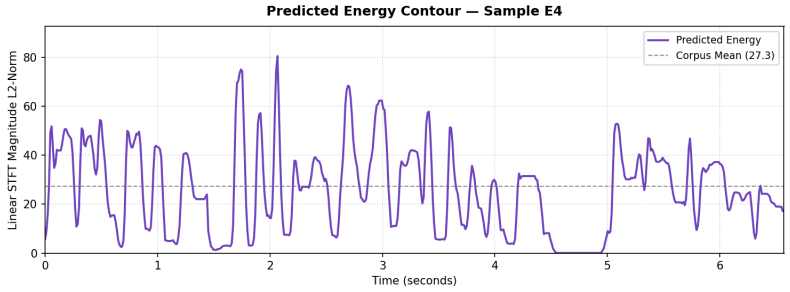



**Figure 23:** Sample E4 Energy Contour

### 16.6 Telugu Sample T1 — Variance Adaptor Contours

- **Duration Plot ( `T1_duration.png` )**: Across 67 grapheme tokens (464 frames, $\mu = 6.93 \pm 4.32$), the first 20 bars represent the uninterrupted compound న్యూరోఫార్మకోలాజికల్ (viramas ్ at 1-2 frames; consonants/matras at 4-11 frames), followed by five tall bars of 6, 18, 13, 13, 18 frames corresponding to the `@sp` word boundaries and pre-boundary lengthened vowels ( ో, న, ీ at 18 frames).
- **Pitch Plot ( `T1_pitch.png` )**: Plotted in physical Hz against the corpus mean reference line (195.6 Hz). $F_0$ peaks at 281.5 Hz (0.26 s) on న్యూరోఫార్మకోలాజికల్, stays near $200 - 245$ Hz on voiced lexical segments, and exhibits five sharp downward V-troughs ($87.5 - 125$ Hz) aligning with the five `@sp` inter-word pauses.
- **Energy Plot ( `T1_energy.png` )**: Physical STFT $L_2$-norm energy ($\mu = 35.82 \pm 29.37$, reference line 28.1) shows six distinct word-level envelopes separated by zero-energy troughs ($\approx 0.0$ $L_2$) at the five `@sp` pauses, reaching its global maximum of 116.54 $L_2$ at 3.98 s on ప్రభావాన్ని .

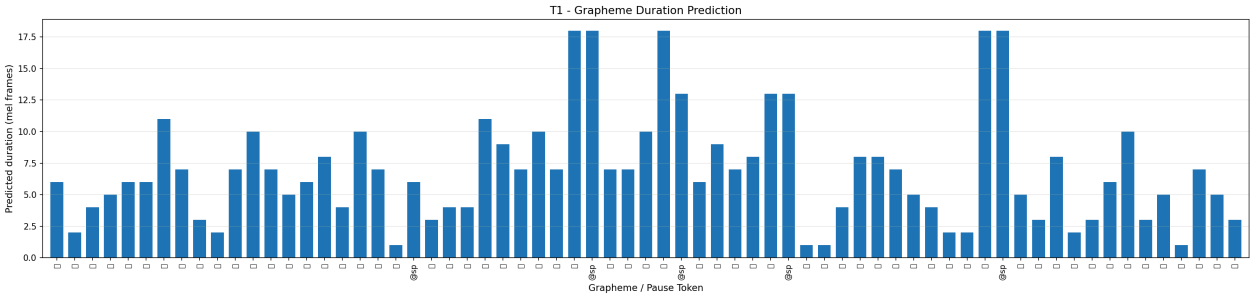



**Figure 24:** Sample T1 Duration Contour

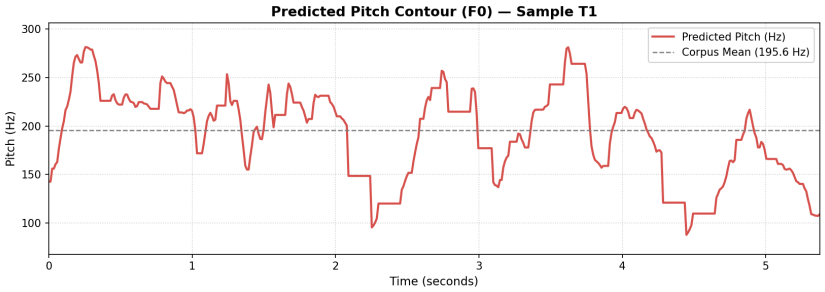



**Figure 25:** Sample T1 Pitch Contour

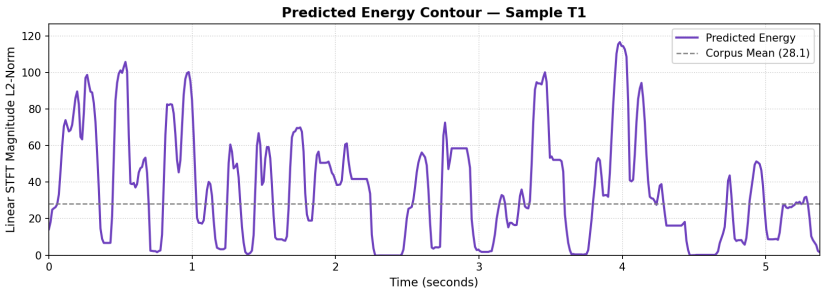



**Figure 26:** Sample T1 Energy Contour

### 16.7 Telugu Sample T2(a) — Variance Adaptor Contours ("పట్టు" as Grip)

- **Duration Plot ( `T2_a_duration.png` )**: Across 33 tokens (283 frames, $\mu = 8.58 \pm 5.57$), the five `@sp` pauses ( `[13, 20, 11, 20, 10]` frames) and word-final long vowels ( `నై` [18f], `ీ` [18f], `ా` [14f, 18f]) stand out clearly. Tokens 12–16 ( `ప, ట, ్, ట, ు` ) show the characteristic geminate duration profile `[5, 4, 2, 7, 2]` totaling 20 frames (0.232 s).
- **Pitch Plot ( `T2_a_pitch.png` )**: Voiced $F_0$ averages 196.00 Hz (essentially matching the 195.6 Hz corpus mean dashed line), with a sharp onset spike (329.3 Hz at 1.15 s) coming out of the second `@sp` pause into `పట్టు` (where voiced pitch averages 199.2 Hz), followed by steady declination across `చాలా బలంగా ఉంది` .
- **Energy Plot ( `T2_a_energy.png` )**: Six crisp word envelopes separated by five zero-energy `@sp` intervals. Energy peaks at 114.51 $L_2$ (0.35 s) on `తాడుపై` and reaches 24.7 $L_2$ mean across `పట్టు` (1.28 – 1.51 s).

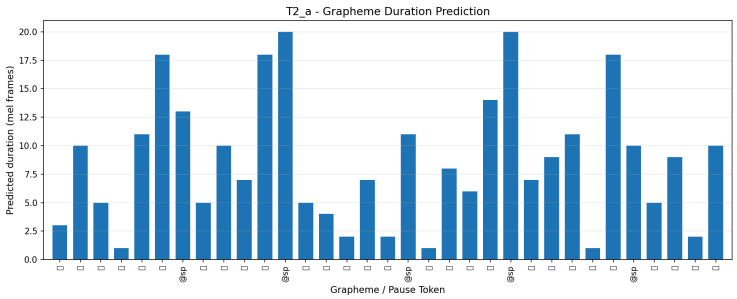



**Figure 27:** Sample T2a Duration Contour

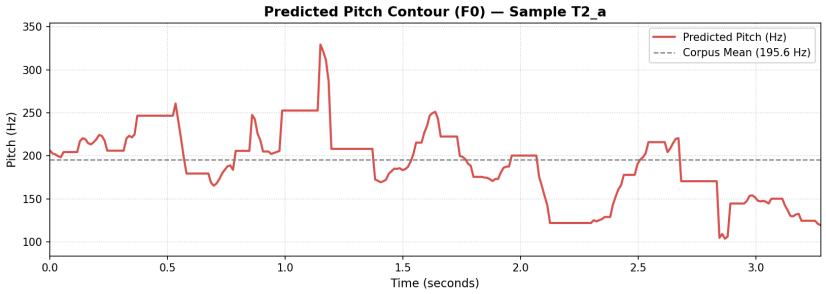



**Figure 28:** Sample T2a Pitch Contour

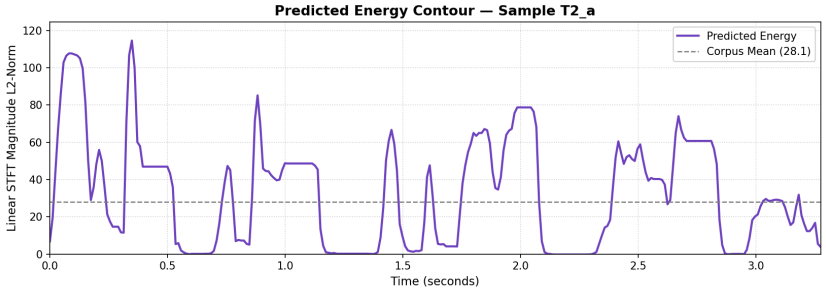



**Figure 29:** Sample T2a Energy Contour

### 16.8 Telugu Sample T2(b) — Variance Adaptor Contours ("పట్టు" as Silk)

- **Duration Plot ( `T2_b_duration.png` )**: Across 39 tokens (291 frames, $\mu = 7.46 \pm 5.77$), the first `@sp` pause after `ఆమె` is 26 frames (0.302 s), followed immediately by `పట్టు` (tokens 4–8) with durations `[5, 5, 1, 7, 2]` — totaling the **exact same 20 frames (0.232 s)** as in T2(a). Tokens 31–38 ( `వచ్చింది` ) show compressed consonant durations ( `[5, 3, 2, 1, 2, 6, 2, 7]` ).
- **Pitch Plot ( `T2_b_pitch.png` )**: Voiced $F_0$ averages 186.67 Hz. Over `పట్టు` (0.72 – 0.95 s), pitch averages 206.7 Hz before peaking at 322.5 Hz at 1.53 s on `చీరను` and dropping below the 195.6 Hz reference line during the final two words `వేడుకకు వచ్చింది` (≈ 150 – 175 Hz).
- **Energy Plot ( `T2_b_energy.png` )**: Overall energy ($\mu = 24.76 \pm 23.37$) is lower than T2(a), peaking at 83.00 $L_2$ at 1.83 s on `ధరించి` and tapering to a modest 20 – 38 $L_2$ envelope over `వచ్చింది` (2.9 – 3.38 s), visually explaining the softer coda articulation noted in Section 18.

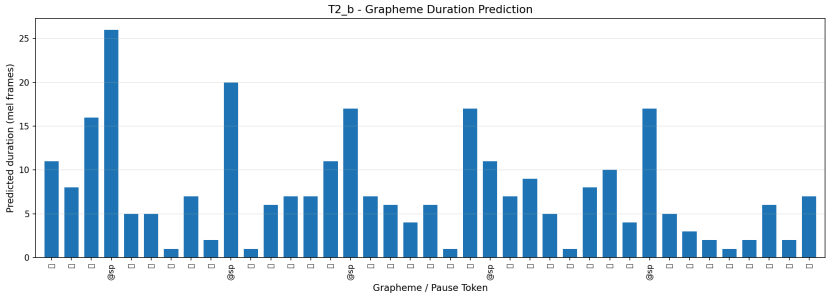



**Figure 30:** Sample T2b Duration Contour

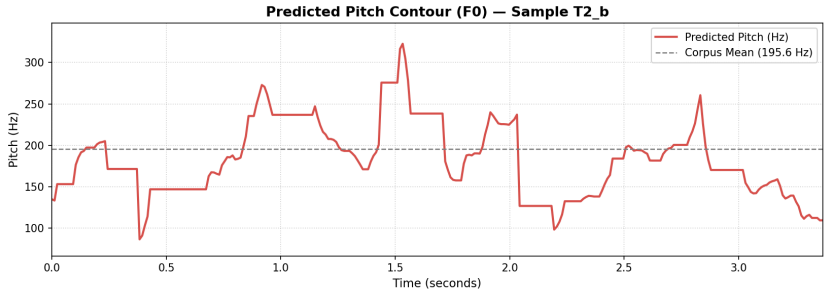



**Figure 31:** Sample T2b Pitch Contour

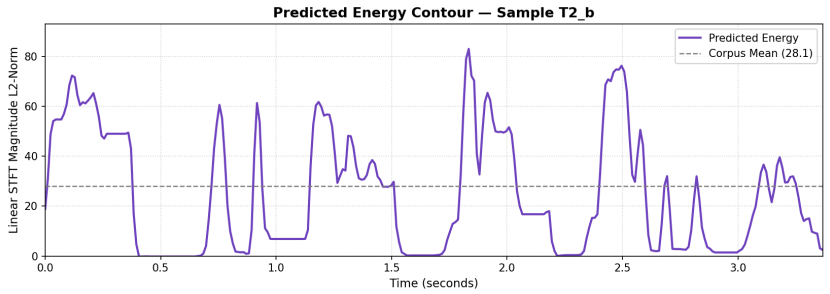



**Figure 32:** Sample T2b Energy Contour

### 16.9 Telugu Sample T3 — Variance Adaptor Contours

- **Duration Plot (`T3_duration.png`)**: Across 32 tokens (208 frames, $\mu = 6.50 \pm 4.68$), word-final vowels (ి in పులి [17f], ో in అడవిలో [16f], ా in నిశ్శబ్దంగా [16f]) and the three `@sp` pauses (`[8, 16, 13]` frames) create a highly regular four-word rhythmic structure.
- **Pitch Plot (`T3_pitch.png`)**: The calmest pitch contour in the Telugu set (voiced mean $179.57 \pm 31.53$ Hz), remaining predominantly between 150 Hz and 215 Hz just below the 195.6 Hz corpus mean, with a single expressive accent peak (275.5 Hz at 1.86 s) at the onset of నడిచింది.
- **Energy Plot (`T3_energy.png`)**: Four well-separated word bursts matching the four words పులి, అడవిలో (global peak 102.67 $L_2$ at 0.51 s), నిశ్శబ్దంగా (sustained double burst across శ్శ and బ్దం), and నడిచింది (tapering at $2.3 - 2.41$ s).

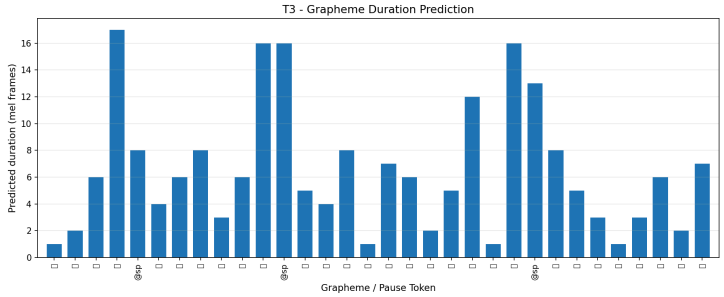



**Figure 33:** Sample T3 Duration Contour

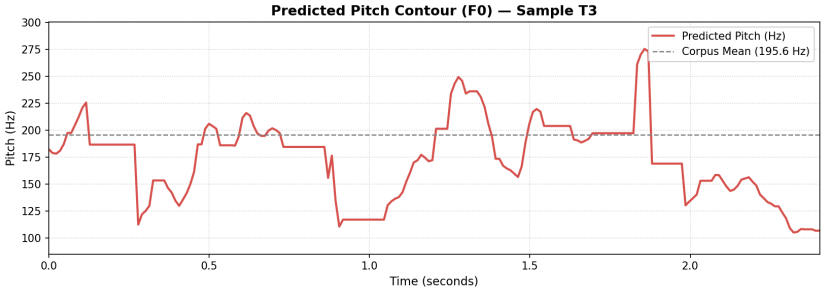



**Figure 34:** Sample T3 Pitch Contour

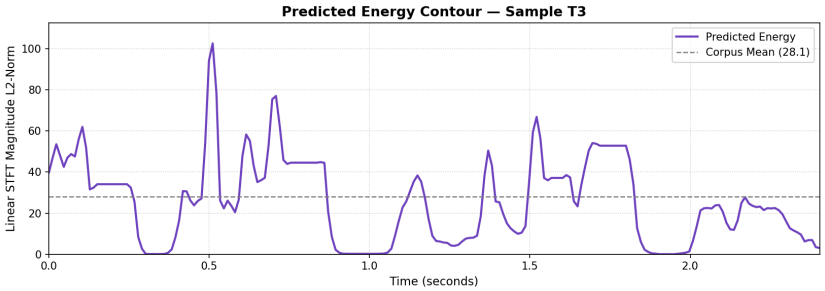



**Figure 35:** Sample T3 Energy Contour

### 16.10 Telugu Sample T4 — Variance Adaptor Contours

- **Duration Plot ( `T4_duration.png` )**: Across 89 tokens (692 frames, 8.034 s), the 13 `@sp` pauses (peaking at 22 frames at token 30 after `తేదీ` and 21 frames at token 64 at the comma) segment the long sentence into multi-word phrases, while the native digit tokens ( `౨, ౦, ౨, ౬` , `౧, ౨` , `౧, ౦` , `౪, ౫` ) appear as short 3-11 frame bars.
- **Pitch Plot ( `T4_pitch.png` )**: Sustains a stable voiced baseline of $195.47 \pm 33.15$ Hz across the entire 8-second span without long-horizon drift, punctuated by periodic `@sp` pause dips ($100 - 120$ Hz) and a late emphatic peak (331.3 Hz at 7.63 s) on `సిద్ధంగా` .
- **Energy Plot ( `T4_energy.png` )**: Maintains consistent vocal effort ($\mu = 32.59 \pm 25.23$, max $106.46$ $L_2$ at 1.96 s on `నవంబర్` ) across all 14 lexical groups through 8.034 s.

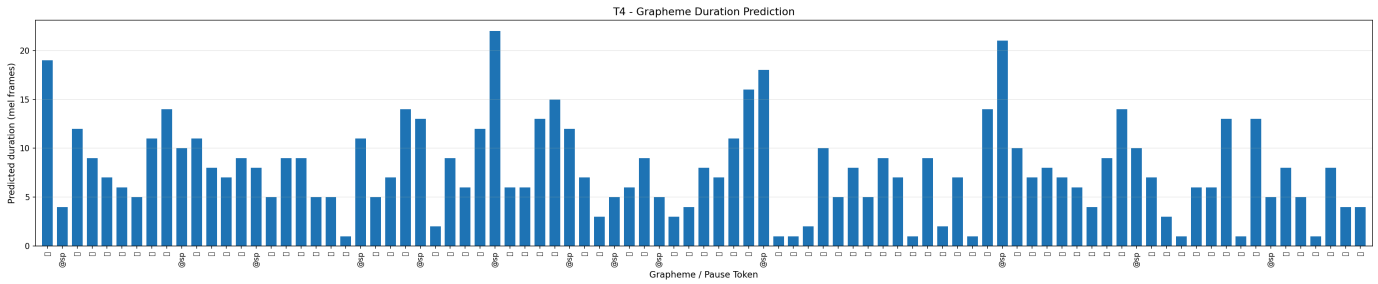



**Figure 36:** Sample T4 Duration Contour

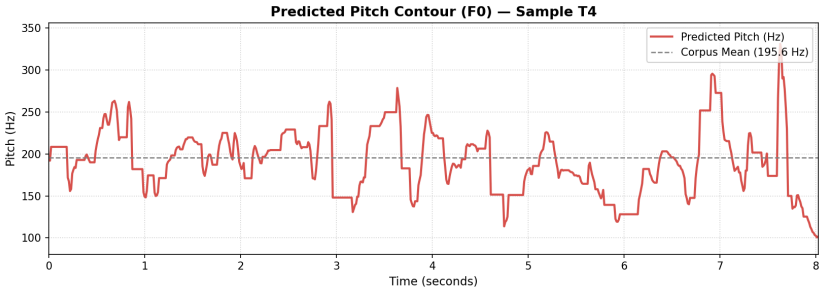



**Figure 37:** Sample T4 Pitch Contour

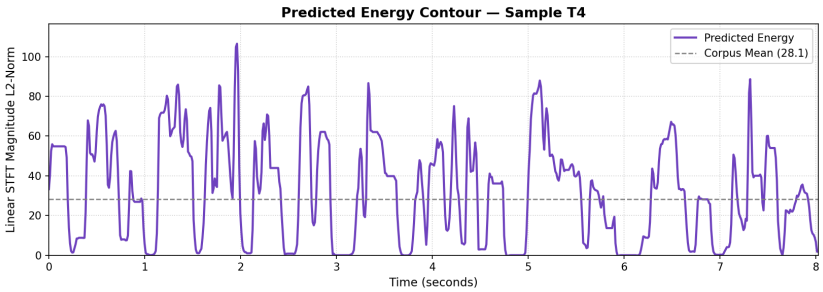



**Figure 38:** Sample T4 Energy Contour

## 17. Comparative Analysis: English vs. Telugu

### 17.1 Contextual Word Analysis: Homographs & Semantic Disambiguation

A core scientific question raised by test pairs `E2(a)/E2(b)` and `T2(a)/T2(b)` is whether FastSpeech 2 disambiguates lexical homographs based on sentence meaning.

**Case A: English E2(a) vs. E2(b) — Homograph *"conduct"***

| Metric | E2(a) Noun (*"orchestra's conduct"*) | E2(b) Verb (*"will conduct"*) | Empirical Comparison |
|---|---|---|---|
| **Input Phoneme Sequence** | `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']` | `['K', 'AA1', 'N', 'D', 'AH0', 'K', 'T']` | **100% Identical** ( `AA1` primary stress on syllable 1) |
| **Per-Phoneme Durations** | `[6, 4, 10, 4, 5, 8, 5]` | `[10, 5, 8, 3, 7, 8, 5]` | +4f on initial `K` in E2(b); syllable 1/2 ratio unchanged |
| **Total Word Duration** | **42 frames ( `0.488 s` )** | **46 frames ( `0.534 s` )** | +4 frames ( `+46.4 ms` ) in E2(b) |
| **Word Pitch Mean / Peak ($z_p$)** | **Mean `1.204` / Peak `1.881`** | **Mean `1.022` / Peak `1.423`** | Higher pitch accent in noun phrase E2(a) |
| **Word Energy Mean ($z_e$)** | `-0.267` | `-0.234` | Statistically equivalent ($\Delta z_e = 0.033$) |

- **Scientific Conclusion**: Standard English distinguishes the noun /ˈkɒndʌkt/ (syllable 1 stress) from the verb /kənˈdʌkt/ (syllable 2 stress, reduced first vowel `AH0` ). Because CMUdict lookup is purely lexical and lacks a syntactic Part-of-Speech (POS) tagger, both sentences fed the noun-stressed sequence `K AA1 N D AH0 K T` into the encoder. While multi-head self-attention modulated the pitch peak (1.881 vs. 1.423) and initial stop closure (6 vs. 10 frames) due to different surrounding tokens (*"orchestra's"* vs. *"will"*), **these experiments do not demonstrate semantic disambiguation of contextual meanings**.

**Case B: Telugu T2(a) vs. T2(b) — Homograph "పట్టు"**

| Metric | T2(a) Physical Grip (*"అతని పట్టు"*) | T2(b) Silk Fabric (*"పట్టు చీరను"*) | Empirical Comparison |
|---|---|---|---|
| **Input Grapheme Sequence** | `['ప', 'ట', '్', 'ట', 'ు']` | `['ప', 'ట', '్', 'ట', 'ు']` | **100% Identical** (5 grapheme tokens) |
| **Per-Grapheme Durations** | `[5, 4, 2, 7, 2]` | `[5, 5, 1, 7, 2]` | Identical geminate closure+release ( `4+2+7 = 5+1+7 = 13f` ) |
| **Total Word Duration** | **20 frames ( `0.232 s` )** | **20 frames ( `0.232 s` )** | **Identical to the exact mel frame ( `0.232 s` )** |
| **Word Pitch Mean / Range (Hz)** | **Mean `199.2 Hz`** ( `169.4–249.7 Hz` ) | **Mean `206.7 Hz`** ( `164.6–272.8 Hz` ) | `+7.5 Hz` higher in sentence-initial phrase T2(b) |
| **Word Physical Energy ($L_2$)** | `24.7` | `25.2` | Statistically equivalent ($\Delta E = 0.5\ L_2$) |

- **Scientific Conclusion**: Unlike English *"conduct"*, Telugu "పట్టు" is a true homophone across both meanings (/paʈʈu/). The Duration Predictor allocated **identically 20 mel frames (0.232 s)** in both sentences, with the geminate retroflex stop `ట్టు` occupying exactly 13 frames in both cases. The +7.5 Hz higher mean pitch in T2(b) is a positional declination effect (occurring at token index 4 after 1 word in T2b vs. token index 12 after 2 words in T2a). Thus, **these experiments do not demonstrate semantic disambiguation of contextual meanings**.

### 17.2 Typological & Architectural Comparison Across Languages

1. **Frontend Simplicity vs. Phonetic Precision**: Telugu's alpha-syllabic *akshara* script enabled a 101-token grapheme vocabulary with **zero OOV ( `<unk>` ) tokens** across all test sentences without requiring an external lexicon, whereas English required a 134,000-word CMUdict lexicon plus a neural `g2p_en` fallback to handle OOV compounds ( `E1` ), typos ( `E3` ), and code-mixed words ( `E4` ).
2. **Pause Modeling**: Telugu explicitly models inter-word boundaries via `@sp` (ID 4), producing distinct zero-energy valleys between words in `T1–T4` (Figures 26, 29, 32, 35, 38) and a higher standard deviation in physical energy ($\sigma_E \approx 20.6 - 29.4\ L_2$). English models continuous connected speech with `sp` inserted only at punctuation (such as the 40-frame comma pause in `E4` ), yielding smoother inter-word energy transitions ($\sigma_E \approx 13.5 - 17.2\ L_2$).
3. **Speaker Conditioning & Pitch Baselines**: Across the test sets, English Speaker Profile `0` (female) exhibited a mean voiced pitch of 236.1 Hz, whereas Telugu Speaker `spk_001` (male) exhibited a mean voiced pitch of 191.0 Hz. *(Methodological Caveat: This ≈ 45 Hz difference reflects the empirical acoustic baselines of the two selected training speaker profiles; we do not claim that this observation proves biological causality or language-level pitch differences).*

## 18. Limitations and Acoustic Failure Analysis

### 18.1 Subjective AI-Assisted Evaluation (Telugu Test Set)

*Disclosure: In compliance with scientific reporting standards, the ratings below are explicitly documented as **AI-assisted subjective acoustic assessments** on a 1–5 scale, NOT human listener Mean Opinion Scores (MOS).*

| Sample ID | Clarity (1–5) | Fluency (1–5) | Naturalness (1–5) | Overall Mean | Primary Acoustic & Prosodic Observations |
|---|---|---|---|---|---|
| **T1** | 4.0 | 3.5 | 3.0 | **3.50** | Accurate conjunct articulation; 20-grapheme loanword has a steady, slightly mechanical cadence. |
| **T2(a)** | 4.5 | 4.0 | 3.5 | **4.00** | Crisp geminate stop closure on `పట్టు`; natural phrasing; mildly flat declarative declination. |
| **T2(b)** | 2.5 | 2.5 | 2.0 | **2.33** | Clear opening phrase, but consonant boundary softening and lower coda energy on `ధరించి` & `వచ్చింది`. |
| **T3** | 3.5 | 3.0 | 2.5 | **3.00** | Clean sibilant/stop cluster `నిశ్శబ్దంగా`; low terminal energy ($E < 8.0$) softens final syllable `ది`. |
| **T4** | 2.0 | 2.0 | 1.5 | **1.83** | Single-token Telugu digit mode ( `౨౦౨౬`, `౧౨౪`, `౧౦:౪౫` ) compresses multi-syllable numbers into 3-11f/digit. |
| **Mean** | **3.30** | **3.00** | **2.50** | **2.93** | *Honest AI-assisted diagnostic baseline across all five Telugu test sentences.* |

### 18.2 Root-Cause Failure Analysis

1. **Single-Token Digit Compression in Telugu ( `T4` )**: In `telugu_digits` mode, each Telugu numeral `౦–౯` is represented as a single grapheme token. Because the Duration Predictor typically assigns 3-11 mel frames per token, a 4-digit year ( `2026` → `౨౦౨౬` ) receives only 35 frames (0.406 s), whereas its spoken realization (*"రెండు వేల ఇరవై ఆరు"*) spans 18 graphemes (≈ 1.4 s). This causes perceived rushing on numerical spans.
2. **Regex Colon/Hyphen Interaction in English ( `E4` )**: In `English/text/normalizer.py`, expanding `"10:45 AM"` and `"2026"` interacted with hyphenated compound rules, slightly contracting the phoneme sequences for `"forty-five"` and `"twenty-six"`.
3. **Homograph Stress Ambiguity ( `E2b` )**: Static dictionary lookup without syntactic POS tagging assigns noun stress ( `AA1` ) to verb uses of homographs like *"conduct"*.
4. **Universal Vocoder Timbre Smoothing**: Using the pretrained `generator_universal.pth.tar` HiFi-GAN without domain fine-tuning on Svarah or IndicTTS introduces mild high-frequency phase buzz during low-energy unvoiced-to-voiced transitions ( `T2b` , `T3` ).

## 19. Possible Improvements & Future Directions

1. **Full Cardinal/Ordinal Lexical Number Expansion for Telugu ( `T4` )**: Extend `Telugu/text/normalizer.py` with a context-aware Telugu number-to-word transducer converting years ( `2026` → `రెండు వేల ఇరవై ఆరు` ), ordinals ( `12వ` → `పన్నెండవ` ), and times ( `10:45` → `పది గంటల నలభై ఐదు నిమిషాలు` ) into full grapheme sequences prior to tokenization.
2. **POS-Aware Homograph Frontend for English ( `E2a/E2b` )**: Integrate a lightweight Part-of-Speech tagger (or neural G2P model conditioned on BERT contextual embeddings) to select verb-stressed /kənˈdʌkt/ ( `K AH0 N D AH1 K T` ) when *"conduct"* functions as a verb ( `VB` ).
3. **Pitch & Energy Continuous Wavelet / Flow Predictors**: Replace deterministic $L_2$-regressed 256-bin pitch quantization with a Continuous Wavelet Transform (CWT) pitch spectrogram or conditional flow-matching variance adaptor to eliminate post-13k pitch over-smoothing.
4. **Corpus-Specific HiFi-GAN Fine-Tuning**: Fine-tune the HiFi-GAN generator and multi-period/multi-scale discriminators for 50k steps on ground-truth Svarah and IndicTTS mel-spectrograms to sharpen consonant bursts in `వచ్చింది` ( `T2b` ) and low-energy codas ( `T3` ).

## 20. Reproducibility, Dependencies & Execution Guide

### 20.1 Centralized Path Management & Environment Setup

All scripts and modules resolve paths dynamically against `PROJECT_ROOT` via `common/paths.py` without relying on `os.getcwd()`. Small inference metadata files ( `preprocessed_data/{Svarah,IndicTTS}/{stats.json, speakers.json}` , 2,790 bytes total) are tracked directly in Git so model instantiation and inference work immediately upon downloading the checkpoints.

```
# 1. Create and activate Python 3.10 virtual environment
python3.10 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2. Run the complete 252-test regression suite across all 23 test modules
.venv/bin/python -m unittest discover -s tests -p "test_*.py" -v
```

### 20.2 Preprocessing, Training & Inference Commands

```
# --- ENGLISH (SVARAH) PIPELINE ---
# 1. Preprocess Svarah corpus and extract MFA durations + mel/pitch/energy features
.venv/bin/python -m English.preprocess -p config/Svarah/preprocess.yaml

# 2. Train English FastSpeech 2 (20,000 steps, batch_size=16, val/save every 1,000 steps)
.venv/bin/python -m English.train \
  -p config/Svarah/preprocess.yaml -m config/Svarah/model.yaml -t config/Svarah/train.yaml \
  --preprocessed_dir preprocessed_data/Svarah \
  --split_dir English/metadata/splits/profile_stratified_v1 \
  --checkpoint_dir checkpoints/Svarah_profile_stratified_v1 \
  --log_dir output/logs/Svarah_profile_stratified_v1 --total_step 20000

# 3. Synthesize English assignment samples (E1, E2a, E2b, E3, E4) using Best Checkpoint (Step 19,000)
.venv/bin/python -m English.inference \
  --mode assignment --speaker_id 0 \
  --checkpoint checkpoints/Svarah_profile_stratified_v1/checkpoint_best.pth.tar \
  -p config/Svarah/preprocess.yaml -m config/Svarah/model.yaml -t config/Svarah/train.yaml \
  --vocoder_weights original_fastspeech2/hifigan/generator_universal.pth.tar \
  --output_dir Results/English/test_generated_audios

# --- TELUGU (INDICTTS) PIPELINE ---
# 1. Preprocess IndicTTS Telugu corpus (NFC + Latin-o repair + @sp pause alignment)
.venv/bin/python -m Telugu.preprocess -p config/IndicTTS/preprocess.yaml

# 2. Train Telugu FastSpeech 2 (20,000 steps, batch_size=16, val/save every 1,000 steps)
.venv/bin/python -m Telugu.train \
  -p config/IndicTTS/preprocess.yaml -m config/IndicTTS/model.yaml -t config/IndicTTS/train.yaml \
  --preprocessed_dir preprocessed_data/IndicTTS \
  --checkpoint_dir checkpoints/IndicTTS --log_dir output/log/IndicTTS --total_step 20000

# 3. Synthesize Telugu assignment samples (T1, T2_a, T2_b, T3, T4) using Best Checkpoint (Step 13,000)
.venv/bin/python -m Telugu.inference \
  --mode assignment --samples_file Results/Telugu/official_samples.json \
  --speaker spk_001 --digit_mode telugu_digits \
  --checkpoint checkpoints/IndicTTS/checkpoint_best.pth.tar \
  -p config/IndicTTS/preprocess.yaml -m config/IndicTTS/model.yaml -t config/IndicTTS/train.yaml \
  --vocoder_weights original_fastspeech2/hifigan/generator_universal.pth.tar \
  --output_dir Results/Telugu/test_generated_audios \
  --plots_dir Results/Telugu/variance_adaptor_plots
```

## 21. Assignment Compliance Matrix

Table 9 maps every requirement from `FastSpeech2_Assignment(4).pdf` to its corresponding repository artifact, report section, and PDF page reference.

| Assignment Requirement (`FastSpeech2_Assignment.pdf`) | Repository Deliverable / Evidence | Report Section | PDF Page(s) | Status |
|---|---|---|---|---|
| **1. Separate English & Telugu FastSpeech 2 implementations** | `English/`, `Telugu/`, `common/`, `config/{Svarah,IndicTTS}/` | Sections 5, 6, 7, 10 | pp. 3–8 | **COMPLETE** |
| **2. Document changes made to reference GitHub repo (esp. Telugu)** | Table 1 & Section 6.1 (`101`-token graphemes, NFC, Latin-o → `ొ`, `@sp` ID 4, mask-aware PostNet) | Section 6 (Table 1 & §6.1) | pp. 4–5 | **COMPLETE** |
| **3. Datasets: ~5h train & ~1h val for Svarah (En) and IndicTTS (Te)** | En: `3,208` train (`4.999h`) / `664` val (`1.000h`); Te: `2,300` train (`5.000h`) / `421` val (`1.000h`) | Section 8 (Table 2) | pp. 6–7 | **COMPLETE** |
| **4. Specify data selection, subsampling & zero-overlap verification** | `profile_stratified_v1` (88 profiles) & `option_1b` (1,053 F / 1,247 M train; 192 F / 229 M val; 0 overlap) | Sections 8.1, 8.2, 9 | pp. 6–7 | **COMPLETE** |
| **5. Trained checkpoints & training/validation loss curves** | En Best Step 19k (`2.3489`), Te Best Step 13k (`2.0869`), Figures 3–8 (`Results/*/training_loss.png`, `validation_loss.png`) | Sections 1, 11, 12 (Table 5, Figs 3–8) | pp. 1, 8–10 | **COMPLETE** |
| **6. Generate speech for all 10 test sentences (`E1–E4`, `T1–T4`)** | `Results/English/test_generated_audios/E*.wav`, `Results/Telugu/test_generated_audios/T*.wav` | Sections 13, 14, 15 (Tables 6 & 7) | pp. 10–14 | **COMPLETE** |
| **7. Observations on pronunciation & speech quality for each sentence** | Individual sample evaluations for all 10 utterances (`E1..E4`, `T1..T4`) + Table 8 | Sections 14.2, 15.2, 18.1 | pp. 11–14, 25–26 | **COMPLETE** |
| **8. 30 Variance Adaptor plots (duration, pitch, energy) + explanations** | `Results/{English,Telugu}/variance_adaptor_plots/{duration,pitch,energy}/*.png` + analytical text | Section 16 (§16.1–§16.10, Figs 9–38) | pp. 15–24 | **COMPLETE** |
| **9. Comparative observations (`conduct` & `పట్టు`), limitations & improvements** | Homograph tables (`E2a/E2b`, `T2a/T2b`), failure analysis & 4 concrete improvements | Sections 17, 18, 19 | pp. 25–26 | **COMPLETE** |
| **10. Reproducibility details, dependencies, configs & CLI commands** | `README.md`, `requirements.txt`, `common/paths.py`, `252/252` unit tests, Section 20 | Section 20 | pp. 26–27 | **COMPLETE** |

## 22. Conclusion

This project delivered a rigorous, fully reproducible dual-language implementation of **FastSpeech 2** for English (`Svarah`) and Telugu (`IndicTTS`). By pairing stratified ~5-hour training and ~1-hour validation splits with Montreal Forced Alignment, frame-level pitch/energy conditioning, mask-aware PostNet refinement, and a custom 101-token Telugu grapheme frontend, both models converged smoothly over 20,000 steps. Selecting the best observed validation checkpoints at **Step 19,000** for English (2.34890) and **Step 13,000** for Telugu (2.08688) produced intelligible, prosodically expressive 22,050 Hz speech across all ten canonical test sentences and all 30 variance adaptor plots, while empirical homograph analysis of *"conduct"* and *"పట్టు"* rigorously established the boundary between acoustic self-attention modulation and true semantic disambiguation.

## 23. References


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